

The Rab7-Epg5 and Rab39-ema modules cooperatively position autophagosomes for efficient lysosomal fusions

Reviewed Preprint

v2 • September 26, 2025

Revised by authors

Reviewed Preprint

v1 • November 7, 2024

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This paper presents **valuable** findings on how autophagosomes are positioned along microtubules for their efficient fusion with lysosomes, providing significant insights into the mechanism. The evidence supporting the conclusions is **solid**, with high-quality fluorescence microscopy combined with *Drosophila* genetics. This work will be of broad interest to cell biologists interested in autophagy and related cell biology fields.

<https://doi.org/10.7554/eLife.102663.2.sa4>

Abstract

Macroautophagy, a major self-degradation pathway in eukaryotic cells, utilizes autophagosomes to transport self-material to lysosomes for degradation. While microtubular transport is crucial for the proper function of autophagy, the exact roles of factors responsible for positioning autophagosomes remain incompletely understood. In this study, we performed a loss-of-function genetic screen targeting genes potentially involved in microtubular motility. A genetic background that blocks autophagosome-lysosome fusions was used to accurately analyze autophagosome positioning. We discovered that pre-fusion autophagosomes move towards the non-centrosomal microtubule organizing center (ncMTOC) in *Drosophila* fat cells, which requires a dynein-dynactin complex. This process is regulated by the small GTPases Rab7 and Rab39 together with their adaptors: Epg5 and ema, respectively. The dynein-dependent movement of vesicles toward the nucleus/ncMTOC is essential for efficient autophagosomal fusions with lysosomes and subsequent degradation. Remarkably, altering the balance of kinesin and dynein motors changes the direction of autophagosome movement, indicating a competitive relationship where normally dynein-mediated transport prevails. Since pre-fusion lysosomes were positioned similarly to

autophagosomes, it indicates that pre-fusion autophagosomes and lysosomes converge at the ncMTOC, which increases the efficiency of vesicle fusions.

Introduction

Macroautophagy is an essential self-degradation pathway in eukaryotic cells, during which double-membrane-bound autophagosomes transport materials to lysosomes for degradation (Parzych & Klionsky, 2014 [↗](#)). Defects in autophagy are associated with multiple pathologies, prompting extensive study of its molecular players over the past decades (Lei & Klionsky, 2021 [↗](#)). During macroautophagy, a double-membrane structure called an autophagosome is formed in an *atg* gene-dependent manner. This autophagosome then fuses with a lysosome or late endosome in a process dependent on the small GTPases Rab7, Rab2, and Arl8, the HOPS tethering complex, and the SNARE proteins Syntaxin17, Ube2A/SNAP29, and Vamp7 (Lőrincz & Juhász, 2020 [↗](#)).

Autophagosomes are suggested to form at random locations within the cytoplasm and are subsequently transported toward the cell center (Jahreiss et al., 2008 [↗](#); Kimura et al., 2008 [↗](#)). Establishing proper proximity between autophagosomes and lysosomes is essential for their ability to fuse. The microtubular network and associated motor proteins are crucial for most vesicular transport. Thus, the involvement of the microtubular system has been suggested in various aspects of autophagosome dynamics, including biogenesis, transport, amphisome formation (Köchl et al., 2006 [↗](#)), and autophagic clearance (Ravikumar et al., 2005 [↗](#)). It is proposed that while microtubules are necessary for the maturation of autophagosomes, their fusion capacity is independent of microtubules (Fass et al., 2006 [↗](#)). Nevertheless, dynein-regulated autophagosomal motility appears indispensable for efficient lysosomal fusion (Jahreiss et al., 2008 [↗](#); Kimura et al., 2008 [↗](#)). How autophagosomes gain the ability to move along microtubules remains unclear. It is suggested that autophagic vesicles acquire dyneins by endosomal fusion (Cheng et al., 2015 [↗](#)); however, autophagosomes still appear to be motile upon Syx17 loss (Neisch et al., 2017 [↗](#)).

Most of our knowledge about autophagosome positioning and movement comes from studies on neurons, where autophagosomes form in the terminal part of axons and then travel toward the soma by dynein-dynactin-regulated bulk retrograde transport during basal autophagy (Ikenaka et al., 2013 [↗](#); Lee et al., 2011 [↗](#); Maday et al., 2012 [↗](#); Wang et al., 2015 [↗](#)). During their route, they fuse with endosomes and lysosomes, resulting in gradual acidification and the acquisition of lysosomal markers (Hill & Colón-Ramos, 2020 [↗](#); Lee et al., 2011 [↗](#)). Degradation takes place in the cell body (Maday et al., 2012 [↗](#)). Inhibited retrograde transport leads to neurodegeneration and defective removal of synaptic autophagosomes (Fu et al., 2014 [↗](#); Lee et al., 2011 [↗](#); Ravikumar et al., 2005 [↗](#)), highlighting the importance of axonal transport in the acidification and degradation processes.

Autophagosomes are suggested to use both dyneins and kinesins in neuronal cells. Initially, they exhibit bidirectional motility at the axon tip, later shifting to dynein-regulated retrograde transport directed toward the soma, where the already mature autolysosomes again show bidirectional motility (Maday et al., 2012 [↗](#)). Various scaffolding proteins have been found to regulate autophagosome transport, including CKA as part of the STRIPAK complex (Neisch et al., 2017 [↗](#)), JIP1, JIP3, and JIP4 (Cason et al., 2021 [↗](#); Cason & Holzbaur, 2023 [↗](#); Fu et al., 2014 [↗](#)), as well as Huntingtin and HAP1 (Cason et al., 2021 [↗](#); Hill & Colón-Ramos, 2020 [↗](#); Wong & Holzbaur, 2014 [↗](#)). It is important to note that before autophagosome closure, phagophores are not able to be transported (Fass et al., 2006 [↗](#)).

However, most of our knowledge about autophagosome motility comes from experimental methods and tools that do not distinguish between non-acidic autophagosomes and autophagic structures that have already undergone some endolysosomal fusion and acidification. For

example, several studies used reporters such as LC3 fused to red fluorescent proteins, which also label post-fusion autolysosomes. Consequently, autophagic organelles that have acidified are sometimes still considered autophagosomes, when in fact they could be autolysosomes.

We and our colleagues have previously identified and characterized several key players in autophagosome-lysosome fusion in starved *Drosophila* fat cells (Boda et al., 2019 [DOI](#); Hegedus et al., 2016 [DOI](#); Lorincz et al., 2017 [DOI](#); Takáts et al., 2013 [DOI](#); Takáts et al., 2014 [DOI](#)). During these studies, we observed an intriguing phenomenon: despite being generated at random positions in the cytosol, autophagosomes accumulated around the nucleus when either the HOPS tethering complex or the SNARE fusion machinery was inhibited (Boda et al., 2019 [DOI](#); Lorincz et al., 2019 [DOI](#); Takáts et al., 2013 [DOI](#); Takáts et al., 2014 [DOI](#)).

This observation led us to employ a novel and unique approach by examining autophagosome positioning in cells where autophagosome-lysosome fusion was inhibited using HOPS RNAi. This method allowed us to exclude the confounding effects of ongoing vesicle fusions, which could otherwise obscure the accurate determination of the roles of different factors in vesicle positioning. These cells were utilized in an RNA interference screen to identify and characterize the molecular participants involved in autophagosome positioning. Our work represents the first comprehensive description of the transport machinery involved in pre-fusion autophagosomes and its significance during autolysosome formation.

Results

We began our investigations by performing an RNA interference screen to identify genes potentially involved in the microtubular positioning of autophagosomes, including MT proteins and motors (such as dynein, dynactin, and kinesin subunits), Rab small GTPases, and their effectors (for the complete list of tested genes and results, see [Supplementary File 1](#) [DOI](#) and [Supplementary Table 1](#) [DOI](#)). We used larval fat cells of the fruit fly (*Drosophila melanogaster*) as a model system, in which bulk macroautophagy was induced by starvation (Scott et al., 2004 [DOI](#)). The fat tissue contained GFP-positive mosaic cells, in which we silenced the gene of interest together with the Vps16A central subunit of the HOPS complex. This RNAi effectively impairs autophagosomal fusion, leading to the accumulation of autophagosomes (Takáts et al., 2014 [DOI](#)). Fat cells also expressed an mCherry-Atg8a reporter driven by a UAS-independent fat body-specific *R4* promoter. This reporter marks both autophagosomes and autolysosomes in control cells, due to the stability of mCherry in acidic environments (**Fig. 1A** [DOI](#)). However, in *vps16a* RNAi cells, which also expressed a control (*luciferase*) RNAi, we observed the accumulation of small mCherry-Atg8a puncta—representing autophagosomes—in the perinuclear region (**Fig. 1A, B, F** [DOI](#)), consistent with previous observations (Takáts et al., 2014 [DOI](#)).

Similar observations were made by endogenous immunostaining against Atg8a (**Figure 1—figure supplement 1A, H** [DOI](#)), Rab7 (**Figure 1—figure supplement 1B, I** [DOI](#)), and Arl8 (**Figure 1—figure supplement 1C, J** [DOI](#)). Endogenous Atg8a immunostaining is specific for autophagosomes (Lorincz et al., 2017 [DOI](#)), Rab7 antibody labels late endosomes, lysosomes, and autophagosomes (Hegedus et al., 2016 [DOI](#)), while Arl8 is a lysosome-specific small GTPase responsible for lysosomal motility and autophagosome-lysosome fusion (Bagshaw et al., 2006 [DOI](#); Boda et al., 2019 [DOI](#); Hofmann & Munro, 2006 [DOI](#)). Our results thus indicate that cells with impaired autophagosome-lysosome fusion accumulate not only pre-fusion autophagosomes but also pre-fusion lysosomes or late endosomes around their nuclei. Although Atg8a signal intensity detected by immunohistochemistry is higher in fusion-incompetent cells than in adjacent control cells due to the accumulation of non-degraded Atg8a (**Figure 1—figure supplement 1A** [DOI](#)), the signal of the mCherry-Atg8a reporter appears slightly weaker in *vps16a* RNAi cells compared to controls (**Fig. 1B** [DOI](#)). This is because the reporter is highly stable and resistant to the acidic environment in autolysosomes (Lorincz et al., 2017 [DOI](#)). To confirm that our mCherry-Atg8a reporter labels structures of autophagic origin, we co-

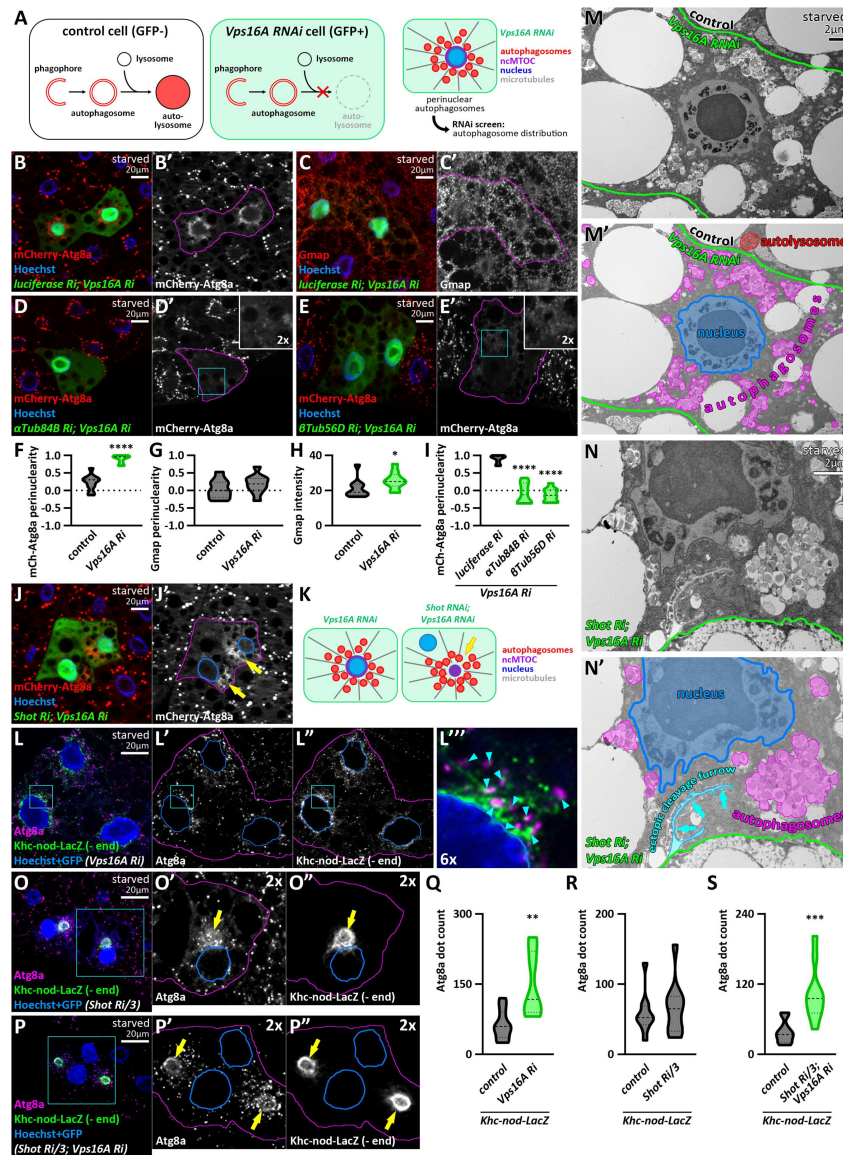


Figure 1.

Autophagosomes move towards the ncMTOC in fat cells.

A: Schematic drawing of the experimental design for screening. **B:** Non-fused autophagosomes accumulate in the perinuclear region upon *Vps16A* silencing. **C:** The cis-Golgi compartment remains unchanged upon the expression of *vps16a* RNAi. **D, E:** The accumulation of autophagosomes is not perinuclear in α or β tubulin; *vps16a* double RNAi cells. The boxed areas in the main panels, marked by cyan, are enlarged in the insets (**D'** and **E'**). **F-I:** Quantification of data shown in **B-E**; $n=10$ cells. **J, K:** Autophagosomes position at an ectopic MTOC (yellow arrows) formed upon *Shot* silencing. (29 out of 41 cells exhibited an ectopic MTOC = 70.73%). **L:** Autophagosomes are near microtubule minus-ends, marked by *Khc-nod-LacZ*. The boxed area in the main panels, marked by cyan, is enlarged in **L'''** (proximity sites indicated by cyan arrowheads). **M, N:** Correlative ultrastructural analysis shows autophagosomes accumulating near the nucleus upon *Vps16A* silencing (border of control and silencing cells marked by green) (**M**). *Shot* knockdown causes aggregation of autophagosomes in ectopic foci in *vps16a* RNAi cells (**N**). An ectopic cleavage furrow (hallmark of *Shot* depletion (Sun et al., 2019)) is also visible (cyan arrows in **N'**). Note: the magnification of **N** is higher to better show this structure. **O, P:** *Atg8a* positive autophagosomes are clustered around the ectopic MTOC (yellow arrows) which is encircled by the signal of the minus-end marker *Khc-nod-LacZ* in *shot* RNAi (**O**) and *vps16a*; *shot* double RNAi cells (**P**). The boxed areas in the main panels, marked by cyan, are enlarged in **O'**, **O''**, **P'** and **P''**. Nuclei are outlined in blue in **J'**, **L'-L'''**, **O'**, **O''**, **P'** and **P''**. The GFP signal of RNAi and *Khc-nod-LacZ* expressing cells is false-colored blue in composite images. **Q-S:** Quantification of data shown in **L, O, P**; $n=10$ cells. The boundaries of RNAi cells are highlighted in magenta in the grayscale panels.

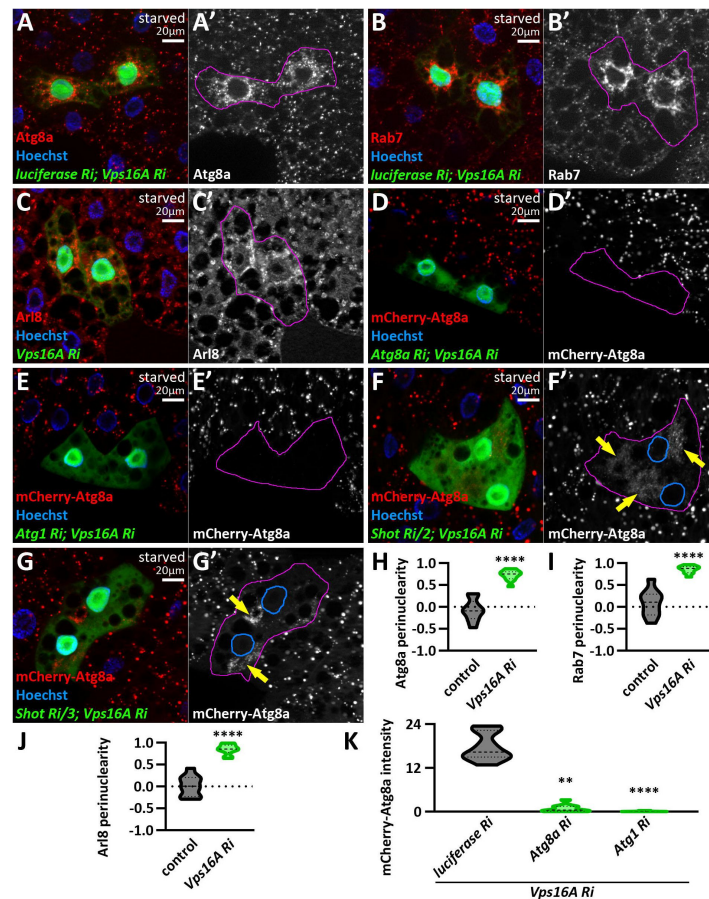


Figure 1—figure supplement 1.

A-C: Immunostainings for Atg8a (A), Rab7 (B), and Arl8 (C) show that pre-fusion autophagosomes (Atg8a and Rab7 positive) and pre-fusion lysosomes (Arl8 positive) accumulate in the perinuclear region in *vps16a* RNAi cells. **D, E:** Atg8a (D) or Atg1 (E) knockdown eliminates the mCherry-Atg8a signal from *vps16a* RNAi cells. **F, G:** Silencing of Shot using two additional independent RNAi lines also results in the formation of an ectopic MTOC (marked by yellow arrows) in *Vps16A* depleted cells, around which autophagosomes accumulate. An ectopic MTOC was present in 29 out of 38 cells (RNAi/2) and 23 out of 29 cells (RNAi/3), representing 76.32% and 79.31%, respectively. **H-K:** Quantification of data shown in A-E; n=10 cells. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. The outlines of nuclei are drawn in blue in F' and G'.

expressed an *atg8a* RNAi with *vps16a* RNAi, which effectively removed the signal of mCherry-Atg8a from the mosaic cells (**Figure 1—figure supplement 1D, K**), confirming that this reporter does not label non-autophagic vacuoles in these cells. Furthermore, the mCherry-Atg8a signal was almost completely absent in *atg1; vps16a* double RNAi cells (**Figure 1—figure supplement 1E, K**), indicating that this reporter is transported to autolysosomes via autophagosomes. To exclude the possibility that the perinuclear accumulation of autophagosomes and unfused lysosomes is due to the overall disorganization of organelles in Vps16A-depleted cells, we immunostained the cells against Gmap, which revealed that the positions of the Golgi apparatuses remained similar to control upon Vps16A silencing. Additionally, the fluorescent signal of Gmap was increased in Vps16A-depleted cells, which is consistent with the fact that the Golgi apparatus is a substrate of golgiphagy in flies (Rahman et al., 2022) (**Fig. 1C, G, H**).

The transport of autophagosomes is microtubule-dependent and minus-end directed

As microtubule (MT)-associated autophagosome transport has been suggested to be more prominent compared to the actomyosin network (Lőrincz & Juhász, 2020), we first silenced the microtubule subunits α - and β -tubulin in *vps16a* RNAi cells to clarify whether the perinuclear localization of autophagosomes is indeed established by the MT network. As expected, knockdown of tubulins diminished the perinuclear localization of autophagosomes and led to their scattering in the cytoplasm (**Fig. 1D, E, I**). Larval fat body cells have been shown to possess a perinuclear, non-centrosomal MTOC (ncMTOC) (Zheng et al., 2020).

This ncMTOC is stabilized by the Spectraplakin Short stop (Shot), and its depletion translocates the perinuclear ncMTOC to an ectopic, cytosolic location (Sun et al., 2019; Zheng et al., 2020). Therefore, we hypothesized that autophagosomes travel towards this MTOC in starved fat cells, and it is the position of the ncMTOC, rather than the nucleus, that determines their direction. Accordingly, autophagosomes accumulated in a central cytosolic region rather than around the nucleus in *shot, vps16a* double RNAi cells (**Fig. 1J, K, Figure 1—figure supplement 1F, G**).

To further confirm that autophagosomes indeed travel towards the ncMTOC, we expressed the MT minus-end reporter Khc-nod-LacZ (a hybrid recombinant kinesin) (Clark et al., 1997) in *vps16a* RNAi cells. Immunolabeling of Atg8a-marked autophagosomes revealed their close proximity to the reporter, which effectively labeled the perinuclear MT network (**Fig. 1L**). Additionally, we performed ultrastructural analysis to further support our findings. Compared to the mostly perinuclear distribution of autophagosomes in Vps16A single knockdown cells (**Fig. 1M**), *shot, vps16a* double RNAi resulted in the concentration of autophagosomes in large groups adjacent to the nucleus, consistent with our fluorescent data (**Fig. 1N**). Moreover, the groups of Atg8a positive autophagic structures observed upon Shot depletion accumulate around a Khc-nod-LacZ positive region, independently of Vps16A (**Fig. 1O-S**). Taken together, our results indicate that autophagosomes move along the MT network oriented towards the MT minus-end to their final destination near the ncMTOC.

A cytoplasmic dynein-dynactin complex transports autophagosomes

Next, we turned to microtubular motor complex subunits. Dynein complexes consist of motor domain-containing heavy chains (HC), intermediate chains (IC), light intermediate chains (LIC), and light chains (LC), and their functions are regulated by dynactin complexes (Canty et al., 2021; Vaughan & Vallee, 1995). The fruit fly genome contains two genes encoding HCs and LICs, one single IC gene, and several genes encoding LCs. These can form several cytoplasmic dynein complexes; our goal was to find the one(s) responsible for autophagosome transport. Upon silencing Dynein heavy chain 64C (Dhc64C, a HC subunit), short wing (sw, an IC subunit), Dynein light intermediate chain (Dlic, a LIC subunit), and roadblock (robl, a LC subunit) in *vps16a*-silenced

cells, we observed an interesting phenomenon: autophagosomes accumulated in the cell periphery, under the plasma membrane, and not around the nucleus (**Fig. 2A–D, K**, **Figure 2—figure supplement 1A, R**). Similar to the dynein hits, the silencing of DCTN1-p150 (Dynactin 1, p150Glued homolog), DCTN2-p50, and DCTN4-p62 also resulted in the redistribution of autophagosomes to the cell periphery (**Fig. 2E–G, K**, **Figure 2—figure supplement 1B, C, R**). Our mCherry-Atg8a data were strengthened by endogenous Atg8a and Rab7 immunostainings (**Figure 2—figure supplement 1D–I, T, U**), as Atg8a and Rab7 positive puncta also redistributed to the cell periphery upon co-silencing Vps16A with our dynein or dynactin hits.

We also overexpressed a dominant-negative form of DCTN1-p150, as well as a wild-type DCTN2-p50/dynamitin, which is described to cause dominant-negative effects when overexpressed (Zheng et al., 2020). These reproduced the phenotypes of dynein or dynactin loss, further strengthening our data (**Figure 2—figure supplement 1J, K, R**). In turn, silencing other dynein or dynactin genes did not cause similar effects; these cells were either control-like (*vps16a* single RNAi) (**Supplementary File 1, Supplementary Table 1**) or, in two cases, a mild scattering of autophagosomes were observed (Dlc90F, an LC subunit, and capping protein alpha, cpa, a dynactin subunit) (**Figure 2—figure supplement 1L, M, S, Supplementary File 1, Supplementary Table 1**).

Our results thus suggest that autophagosomes are mainly transported by a cytosolic dynein complex composed of Dhc64 (HC), sw (IC), Dlic (LIC), and roadblock (LC), regulated by a DCTN1-2-4 containing dynactin complex (**Fig. 2J**). However, we cannot exclude the possibility that other dynein and dynactin subunits also contribute to autophagosome motility as this would require confirmation that the RNAi transgenes yielding negative results were indeed efficient in generating loss-of-function of their target genes. Importantly, the peripheral accumulation of autophagosomes upon the lack of dynein-regulated movement suggests that kinesins can take their place and carry autophagosomes to the positive end of microtubules.

We continued screening by silencing dynein activators and regulators. These proteins, including the Bicaudal-D, Hook, and Ninein families in mammals, are responsible for enhancing processive motility and recruiting cargo to the dynein-dynactin complex (Olenick & Holzbaur, 2019; Redwine et al., 2017). We found that the silencing of a candidate activator, Girdin (Redwine et al., 2017), in a *vps16a* RNAi background led to the dispersal of autophagosomes (**Figure 2—figure supplement 1N, V**), similarly to the loss of Lis-1, a well-studied and essential regulator of dynein motor function (Dix et al., 2013; Siller et al., 2005; Sitaram et al., 2012; Swan et al., 1999) (**Figure 2—figure supplement 1O, V**). This can be explained by the fact that some dynein function is still present without Girdin and Lis-1, and their loss does not completely abolish dynein activity.

Therefore, next our screen focused on kinesin motors. Importantly, none of the kinesin knockdowns inhibited or significantly enhanced the perinuclear positioning of autophagosomes in *vps16a*-depleted cells (Khc and Klc as examples are shown, **Fig. 2H, I, K**, **Figure 2—figure supplement 1P, Q, T, Supplementary File 1, Supplementary Table 1**). These results suggest that cells predominantly use the dynein complex to transport autophagosomes, rather than kinesins.

A proper dynein/kinesin ratio determines the direction of autophagosome positioning

Since dynein loss leads to the peripheral relocation of autophagosomes, we hypothesized that in this case, kinesins take over the role of transporting autophagosomes, preferring the opposite direction. To examine this possibility, we overexpressed two kinesin motors (Klp67A, Klp98A) in *vps16a* RNAi cells. Strikingly, both resulted in the scattering of autophagosomes, and in some cases, caused the peripheral accumulation of mCherry-Atg8a puncta, resembling dynein loss (**Fig. 3A, B**,

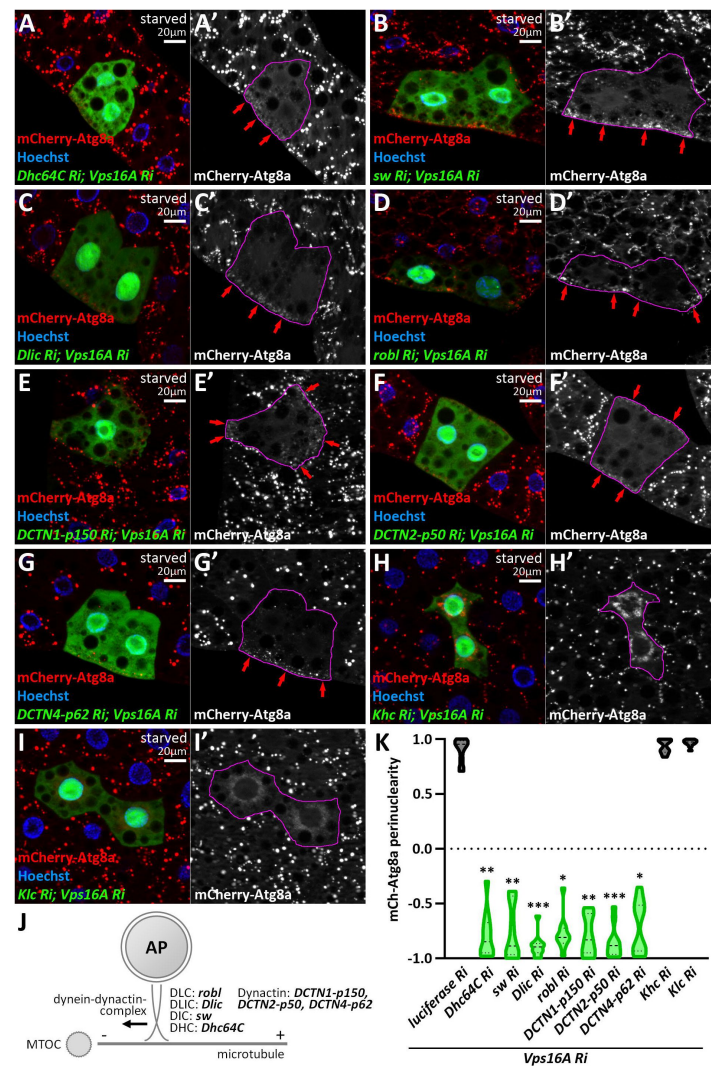


Figure 2.

A dynein-dynactin complex is required for minus-end directed autophagosome transport.

A-G: Knockdown of dynein (A-D) and dynactin subunits (E-G) results in the peripheral redistribution of autophagosomes in *vps16a* RNAi cells (red arrows). **H, I:** Kinesin silencing does not affect the perinuclear accumulation of autophagosomes in *vps16a* RNAi cells. **J:** Proposed model of the suggested dynein-dynactin complex responsible for autophagosome positioning in fat cells. DHC: dynein heavy chain; DIC: dynein intermediate chain; DLIC: dynein light intermediate chain; DLC: dynein light chain. **K:** Quantification of data shown in A-I; n=10 cells. The boundaries of RNAi cells are highlighted in magenta in the grayscale panels.

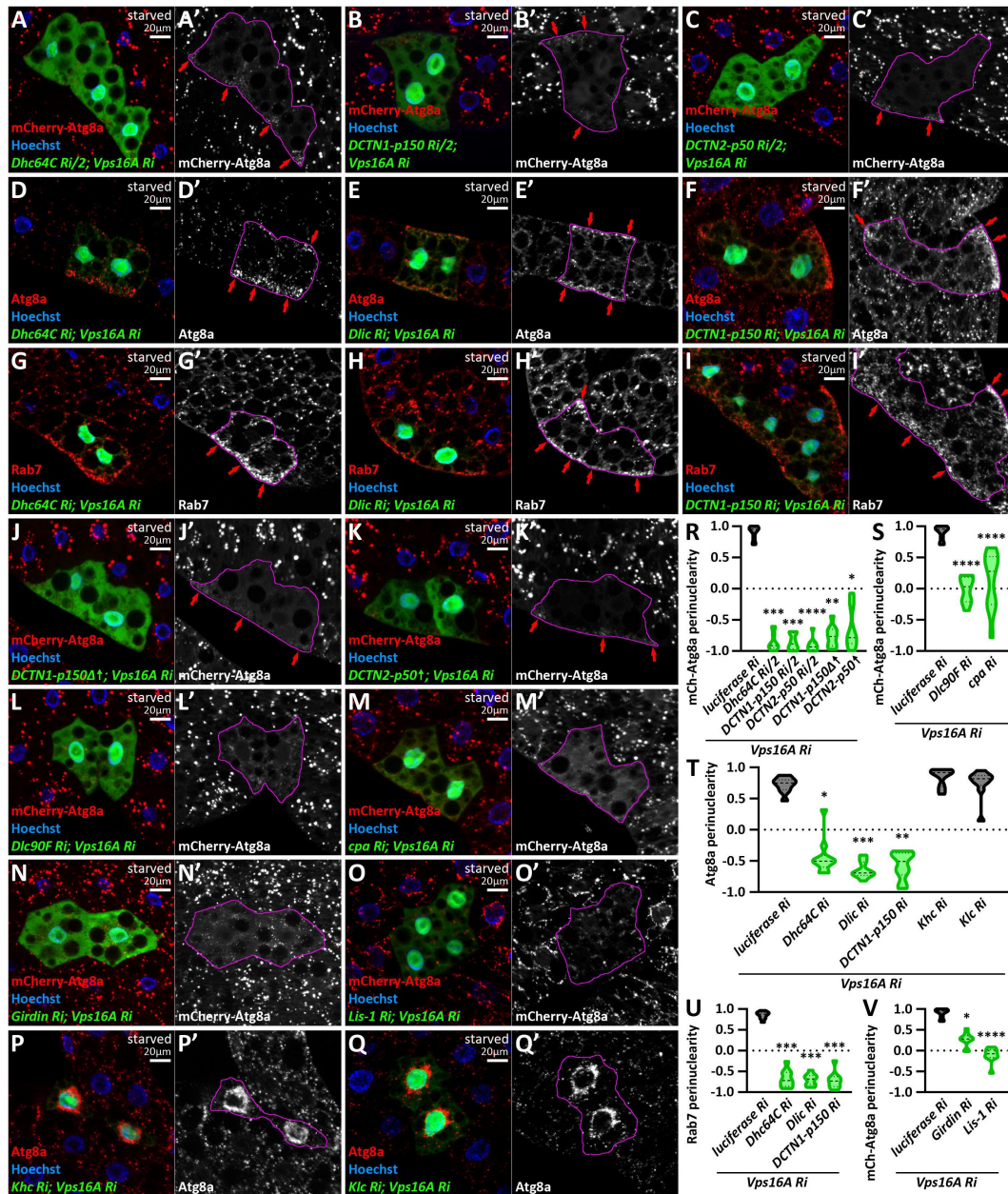


Figure 2—figure supplement 1.

A-C: Silencing of dynein and dynactin subunits (*Dhc64C* in A, *DCTN1-p150* in B, and *DCTN2-p50* in C) using independent RNAi lines also leads to the peripheral redistribution of mCherry-Atg8a positive autophagosomes in *vps16a* RNAi cells (red arrows). **D-I:** Peripheral accumulation (red arrows) of Atg8a (D-F) or Rab7 (G-I) positive autophagosomes can be seen in *dhc64c*, *vps16a* (D, G), *dlc*, *vps16a* (E, H), or *dctn1-p150*, *vps16a* (F, I) double RNAi cells. **J, K:** Overexpression of a dominant negative form of *DCTN1-p150* or wild type *DCTN2-p50* (which exhibits a dominant-negative effect) in *Vps16A* depleted cells also results in the peripheral redistribution of mCherry-Atg8a positive autophagosomes (red arrows). **L, M:** Silencing of the dynein light chain *Dlc90F* and the dynactin subunit *cpa* in *Vps16A* KD cells results in a weaker phenotype, with mCherry-Atg8a positive autophagosomes scattered in the cytosol. **N, O:** Knockdown of *Girdin* (N), a suggested dynein activator, and the dynein regulator *Lis-1* (O) results in a generally dispersed distribution of mCherry-Atg8a positive autophagosomes in *vps16a* RNAi cells. **P, Q:** The perinuclear distribution of autophagosomes in *vps16a* RNAi cells remains unaffected by the co-expression of *khc* (P) or *klc* (Q) RNAi. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. **R-V:** Quantification of data shown in A-Q; n=10 cells.

F [↗](#)). Moreover, autophagosomes scattered upon co-silencing a dynactin and a kinesin in *vps16a* RNAi cells (**Fig. 3C**, **F** [↗](#)). This suggests that without MT motors, autophagosomes are unable to move properly.

To further examine the relationship between dyneins and kinesins, we expressed the recombinant kinesin Khc-nod-LacZ in *vps16a* single and dynactin, *vps16a* double RNAi cells. This recombinant kinesin contains the cargo domain of Khc but moves towards the MT minus-end, similar to dyneins. Interestingly, overexpression of Khc-nod-LacZ in *vps16a* single RNAi cells appeared to influence the assembly and/or function of the ncMTOC, as the nuclear envelope was only partly surrounded by autophagosomes, but their distribution remained perinuclear (**Fig. 3D**, **F** [↗](#)). Strikingly, when we overexpressed this recombinant kinesin in dynactin, *Vps16A* double knockdown cells, autophagosomes became dispersed (**Fig. 3E**, **F** [↗](#)), and no longer accumulated at the periphery as seen in *dynactin1-p150*, *vps16a* double RNAi cells. Since kinesins do not require the activity of dynactins, this suggests that this recombinant kinesin could partially rescue the dynein/dynactin function loss and take over the role of the missing minus-end motors, further supporting that a proper dynein/kinesin ratio determines the direction of autophagosome positioning.

Since peripheral autophagosome accumulation upon loss of dyneins and dynactins tended to occur on the side of marginal fat body cells facing the body cavity and in contact with the blood (**Fig. 2A-G** [↗](#)), we asked whether kinesins also follow a similar distribution pattern. To examine this, we performed anti-HA immunostaining on HA-tagged Klp98A-overexpressing cells and found that this reporter is indeed enriched at the fat body margin (**Fig. 3G**, **H** [↗](#)), supporting the hypothesis that the autophagosome redistribution upon loss of minus end transport is kinesin-dependent.

Rab7 and Rab39 and their effectors Epg5 and ema are required for bidirectional autophagosome transport

Rab small GTPases regulate vesicle transport and fusion by recruiting different effectors in their active, GTP-bound form (Stenmark, 2009 [↗](#)). Therefore, we screened all the Rab small GTPases. In Rab7, or subunits of its guanine nucleotide exchange factor complex (Mon1-Ccz1) and its interactor Epg5 (Gillingham et al., 2014 [↗](#)) knockdown cells with simultaneous *Vps16A* silencing, we could no longer observe the perinuclear accumulation of autophagosomes; they were scattered throughout the cytoplasm (**Fig. 4A-D**, **J** [↗](#)). Other hits were Rab39 and its interactor ema (Gillingham et al., 2014 [↗](#)), both suggested to be involved in the regulation of lysosomal degradation (Kim et al., 2012 [↗](#); Kim et al., 2010 [↗](#); Lakatos et al., 2021 [↗](#); Zhang et al., 2023 [↗](#)). Their phenotype was very similar (**Fig. 4E**, **F**, **J** [↗](#), **Figure 4—figure supplement 1A**, **G** [↗](#)) to the loss of Rab7 and Epg5.

Importantly, no other interaction partners of Rab7 and Rab39 appeared to be required for autophagosome transport (**Supplementary File 1, Supplementary Table 1** [↗](#); examples shown in **Fig. 4G**, **H**, **J** [↗](#)), suggesting that mainly the Rab7-Epg5 and Rab39-ema interactions are required for the bidirectional motility of autophagosomes (**Fig. 4I** [↗](#)). Our results were also strengthened by immunolabelings of Atg8a and Rab7 (**Fig. 5A**, **B**, **E**, **F** [↗](#), **Figure 4—figure supplement 1B-F**, **H**, **I** [↗](#)). Importantly, autophagosomes were still dispersed in *epg5*, *vps16a* double RNAi cells, even if we overexpressed a YFP-Rab7 transgene, suggesting that Rab7 indeed regulates autophagosome positioning via Epg5 (**Fig. 5C**, **G** [↗](#)). Interestingly, Arl8 immunolabeling revealed that Epg5 loss does not influence the perinuclear positioning of non-fused lysosomes (**Fig. 5D**, **H** [↗](#)), suggesting that it exerts its function on autophagosomes.

Therefore, we further analyzed Epg5 functions. We first generated an *Epg5-9xHA* transgene driven by the *epg5* genomic promoter and expressed this reporter in *Drosophila* S2R+ cells. We found that this reporter colocalizes with both endogenous Rab7 and Atg8a (**Fig. 5I**, **J** [↗](#)). Moreover, we

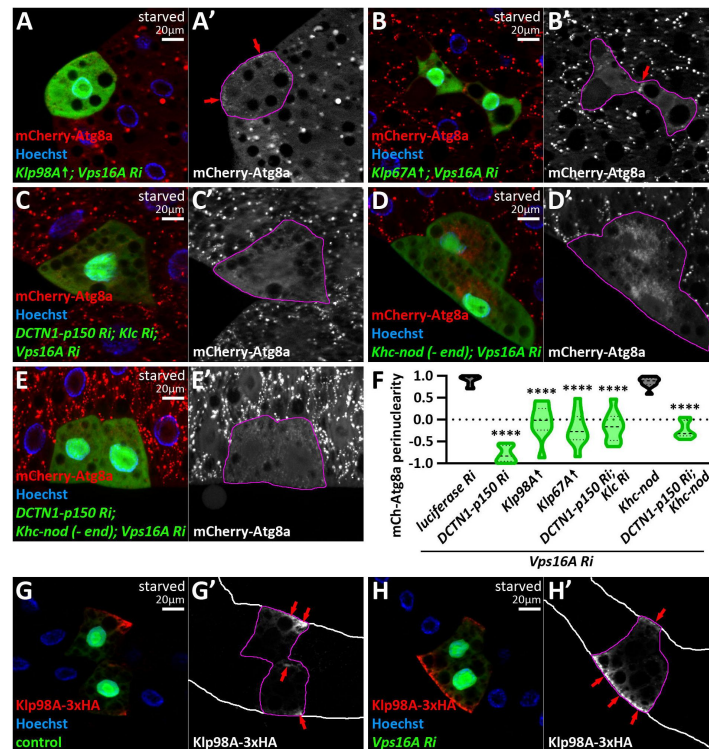


Figure 3.

The proper dynein/kinesin ratio determines the directionality of autophagosome transport.

A, B: Overexpression of kinesin motors blocks the minus-end transport of autophagosomes, leading to their accumulation in the cell periphery in *vps16a* RNAi cells (red arrows). **C:** In dynactin-kinesin-*vps16a* triple RNAi cells, autophagosomes are distributed throughout the cytoplasm. **D:** Overexpression of the recombinant minus-end motor Khc-nod-LacZ does not affect minus-end directed autophagosome transport in *vps16a* RNAi cells. **E:** Khc-nod-LacZ expression partially rescues the dynactin KD-induced peripheral redistribution of autophagosomes in *vps16a* RNAi cells. **F:** Quantification of data shown in A-E; n=10. Data for *dctn1-p150* RNAi is included as a positive control for peripheral distribution (shown in **Figure 2E**, K). **G, H:** Overexpressed Klp98A-3xHA accumulates at the periphery of marginal fat body cells, specifically on the side facing the body cavity and in contact with the hemolymph (red arrows), in both control (G) and fusion-inhibited (*vps16a* RNAi) cells (H). The boundaries of RNAi or kinesin overexpressing cells are highlighted in magenta in the grayscale panels. Fat body edges are outlined in white in G' and H'.

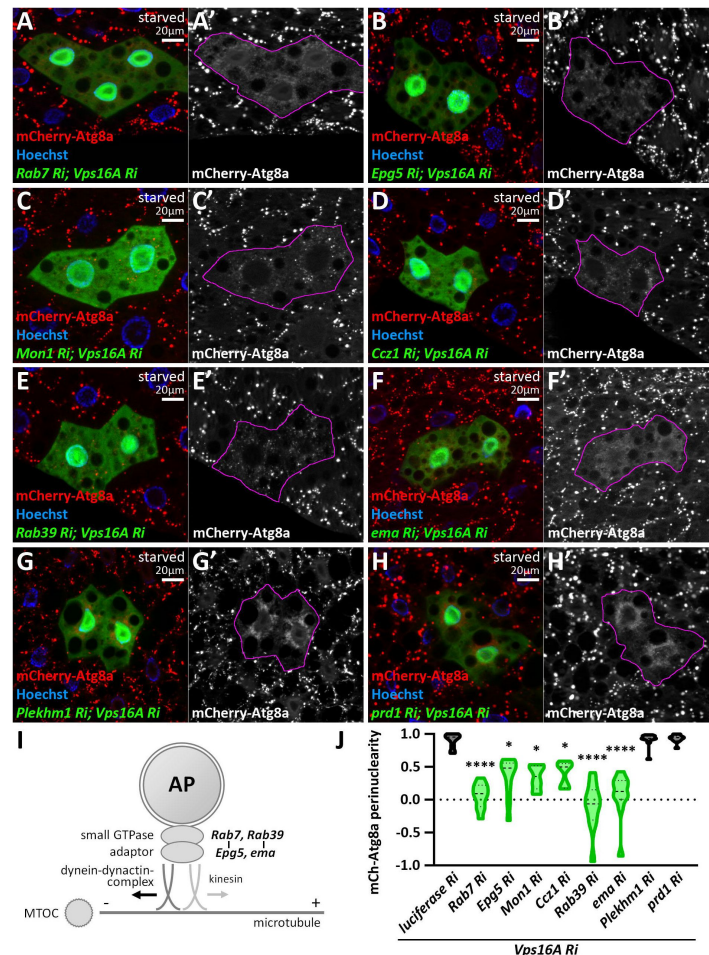


Figure 4.

Rab7 and Rab39 small GTPases and their interactors are responsible for bidirectional movement of autophagosomes.

A-H: Knockdown of Rab7 (A), its interactor Epg5 (B), the subunits of its guanine nucleotide exchange factor Mon1 (C) and Ccz1 (D), as well as Rab39 (E) and its interactor ema (F), inhibits the perinuclear positioning of autophagosomes in *vps16A* RNAi cells. In contrast, other factors such as Plekhm1 (G) and prd1 (H) do not affect autophagosome positioning. **I:** Proposed model of Rab small GTPases with their adaptors involved in autophagosome positioning. **J:** Quantification of data shown in A-H; n=10 cells. The boundaries of RNAi cells are highlighted in magenta in the grayscale panels.

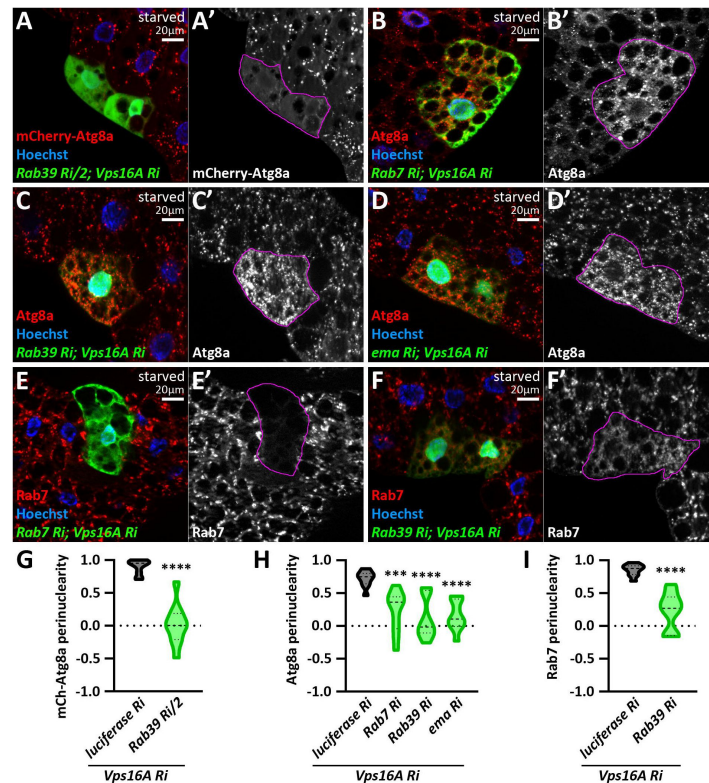


Figure 4—figure supplement 1.

A: Expression of an independent *rab39* RNAi in *vps16a* RNAi cells results in the scattering of mCherry-Atg8a positive autophagosomes. **B:** Atg8a positive autophagosomes accumulate throughout the cytoplasm in *rab7*, *vps16a* double RNAi cells. **C-D:** Atg8a positive autophagosomes are scattered throughout the cytosol in *rab39*, *vps16a* (C) or *ema*, *vps16a* (D) double RNAi cells. **E:** Expression of *rab7* RNAi in *Vps16A* KD cells eliminates the Rab7 signal, indicating effective RNA interference. **F:** Rab7 positive structures (autophagosomes) are scattered throughout the cytosol in *rab39*, *vps16a* double RNAi cells. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. **G-I:** Quantification of data shown in A-D, F; n=10 cells.

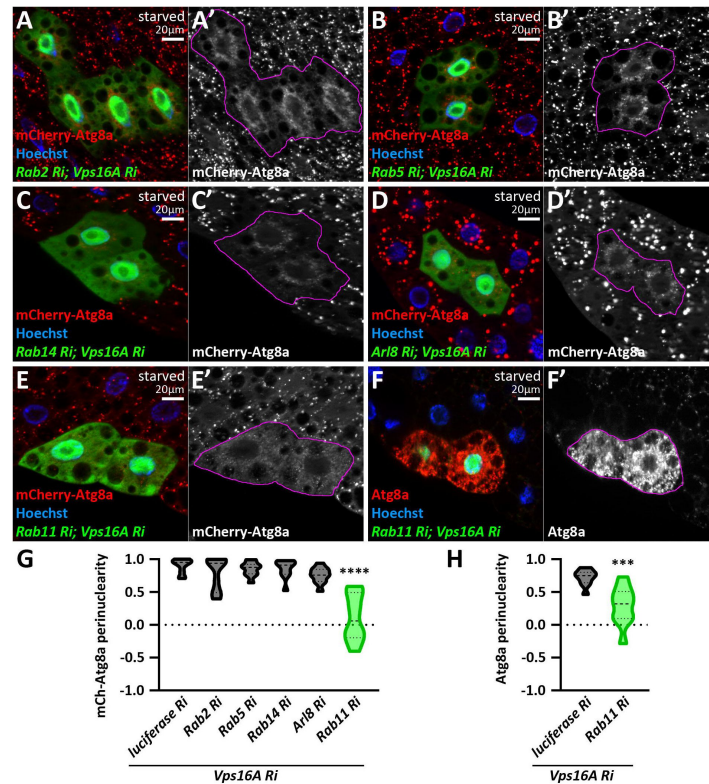


Figure 4—figure supplement 2.

A-D: Silencing the endolysosomal small GTPases Rab2 (A), Rab5 (B), Rab14 (C), and Arl8 (D) does not alter the perinuclear positioning of mCherry-Atg8a positive autophagosomes in *vps16A* RNAi cells. **E, F:** Rab11 knockdown leads to the scattering of mCherry-Atg8a positive autophagosomes in *vps16A* RNAi cells (E), which appears to be due to the extreme accumulation of autophagosomes, as indicated by endogenous Atg8a immunostaining (F). The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. **G, H:** Quantification of data shown in A-F; n=10 cells.

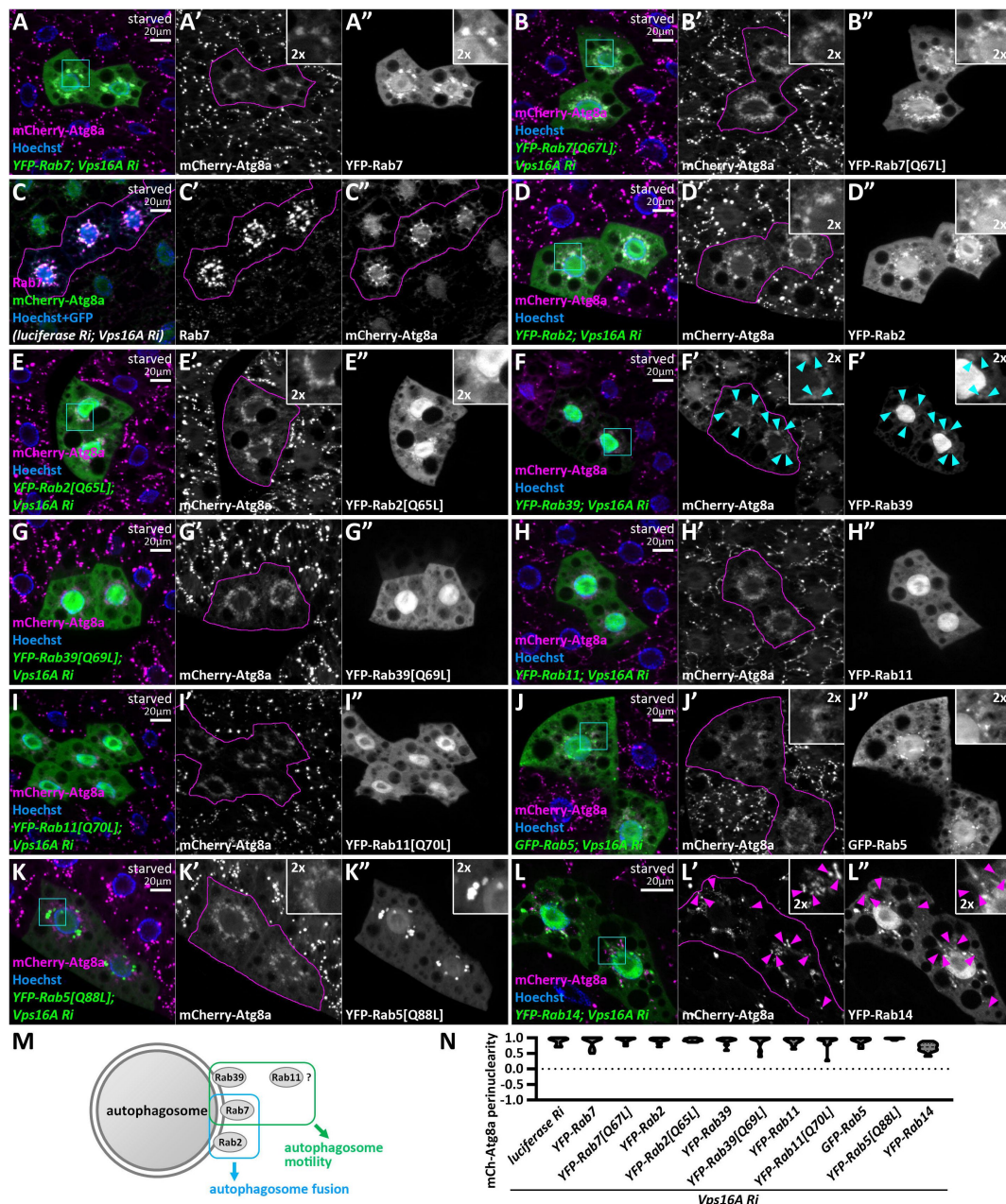


Figure 4—figure supplement 3.

A-L: Overexpressing the wild-type (WT) and constitutively active (CA) YFP or GFP-tagged forms of selected endolysosomal/autophagosomal Rab GTPases (Rab7-WT: A, Rab7-CA: B, Rab2-WT: D, Rab2-CA: E, Rab39-WT: F, Rab39-CA: G, Rab11-WT: H, Rab11-CA: I, Rab5-WT: J, Rab5-CA: K, Rab14-WT: L) had no effect on the perinuclear distribution of mCherry-Atg8a positive autophagosomes in *vps16a* RNAi cells. The signals of both WT and CA forms of YFP-Rab7 and YFP-Rab2 strongly overlapped with mCherry-Atg8a positive autophagosomes in *vps16a* RNAi cells (A, B, D, E), similar to endogenous Rab7 as evidenced by Rab7 immunostaining (C). The GFP signal of RNAi-expressing cells is false-colored blue in C. In contrast, the YFP-tagged WT form of Rab39 showed weak but apparent colocalization (cyan arrowheads) (F), while Rab11 did not colocalize with mCherry-Atg8a even when its CA form was expressed (H, I). Neither form of Rab5 overlapped with mCherry-Atg8a puncta (J, K), similar to Rab14 (L), indicating that these structures are either endosomes or lysosomes (magenta arrowheads point to autophagosomes in the grayscale panels of L). The boundaries of YFP/GFP-tagged Rab or GFP expressing cells are highlighted in magenta in the grayscale panels showing mCherry-Atg8a or Rab channels. The boxed areas in the main panels, marked by cyan, are enlarged in the insets. **M:** Potential roles of Rab GTPases in autophagosome transport and fusion. **N:** Quantification of data shown in A, B, D-L; n=10 cells.

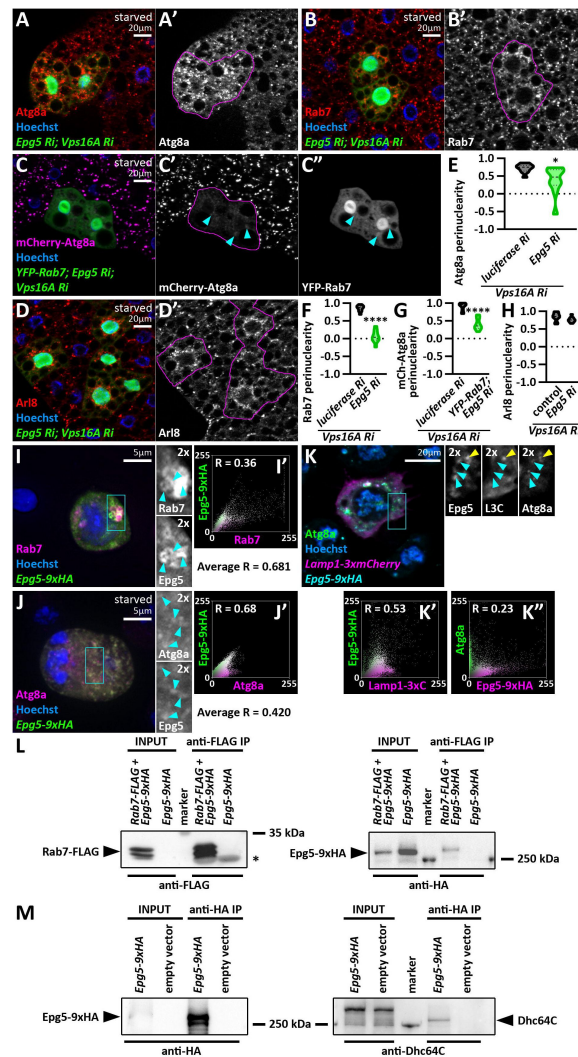


Figure 5.

Epg5 is responsible for bidirectional movement of autophagosomes.

A-B: The distribution of Atg8a (A) or Rab7 (B) positive autophagosomes becomes dispersed upon the expression of *epg5* and *vps16a* RNAi. **C:** Overexpression of YFP-tagged Rab7 does not rescue the scattered distribution of mCherry-Atg8a positive autophagosomes in the absence of Epg5 in *vps16a* RNAi cells, even though the colocalization of YFP-Rab7 with mCherry-Atg8a remains unaffected. Cyan arrowheads in the grayscale panels point to YFP-Rab7 and mCherry-Atg8a double positive dots. **D:** The localization of Arl8 positive lysosomes remains perinuclear in *epg5*, *vps16a* double RNAi cells. The boundaries of RNAi and YFP-Rab7 expressing cells are highlighted in magenta in the grayscale panels. **E-H:** Quantification of data shown in A-D; n=10 cells. **I, J:** Epg5-9xHA colocalizes with endogenous Rab7 (I) or Atg8a (J) positive structures in S2R+ cells. Cyan arrowheads within insets (marked by cyan boxes in panels I and J) point to Epg5-9xHA and Rab7 or Atg8a double positive structures, respectively. **I'** and **J'** show scatter plots generated from the images of cells in panels I and J, respectively, depicting the intensity correlation profiles of Epg5-9xHA with Rab7 or Atg8a. Pearson correlation coefficients (R) are indicated, with the average R (n=10 cells) also shown, indicating colocalization in both cases. **K:** Epg5-9xHA colocalizes with Atg8a positive, Lamp1-3xmCherry negative (pre-fusion) autophagosomes, as well as with Atg8a and Lamp1-3xmCherry double positive autolysosomes in S2R+ cells. Cyan arrowheads in insets (marked by a cyan box in panel K) point to Epg5-9xHA, Atg8a double positive, Lamp1-3xmCherry negative structures, while a yellow arrowhead marks a triple positive autolysosome. **K'** and **K'':** Scatter plots based on the cell in panel K, show intensity correlations of Epg5-9xHA with Lamp1-3xmCherry and Atg8a, respectively. Pearson correlation coefficients indicate partial colocalizations. **L, M:** Coimmunoprecipitation experiments show that Epg5-9xHA binds to Rab7-FLAG (L) and endogenous Dhc64C (M) in cultured *Drosophila* cells. The asterisk in L marks immunoglobulin light chain. The smeared input bands of Dhc64C in panel M are due to the large size of Dhc64C, which affects its migration characteristics.

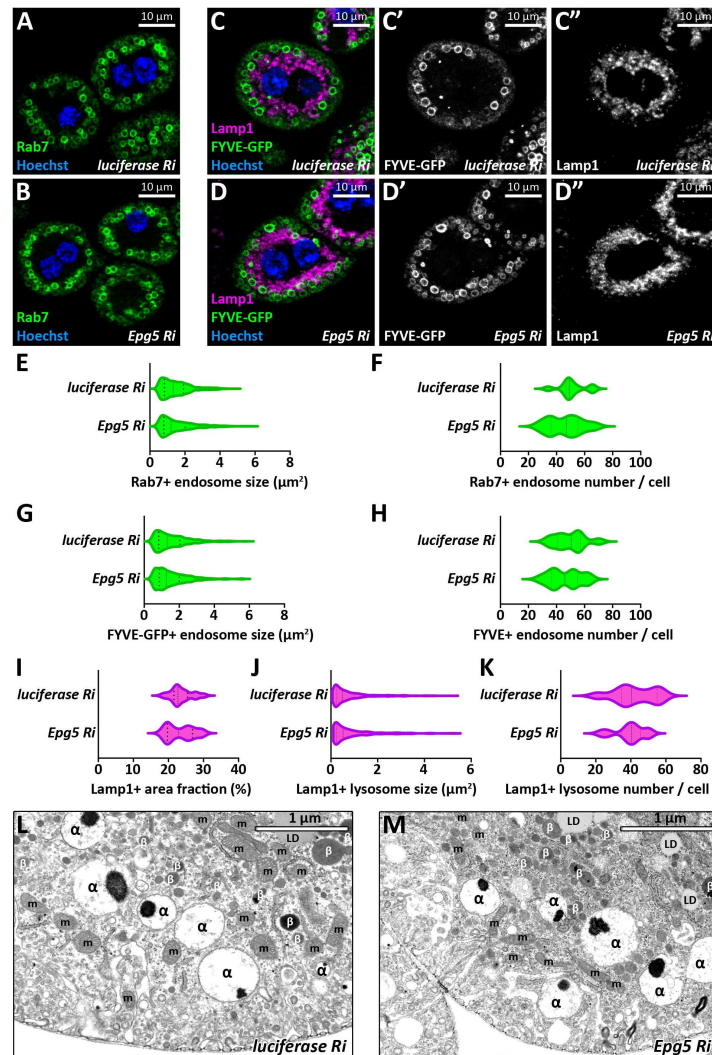


Figure 5—figure supplement 1.

A-D: The size, number and distribution of Rab7 (A, B) or FYVE-GFP positive endosomes (C, D) and Lamp1 positive lysosomes (C, D) remain unaffected by *epg5* RNAi expression (B, D) in nephrocytes, compared to controls (A, C). **E-K:** Quantification of data shown in A-D; E: n=512 (control) and 461 (*epg5* RNAi) endosomes from 10 cells. F: n=10 cells. G: n=496 (control) and 461 (*epg5* RNAi) endosomes from 10 cells. H: n=10 cells. I: n=15 cells. J: n=422 (control) and 397 (*epg5* RNAi) lysosomes from 10 cells. K: n=10 cells. **L, M:** Ultrastructural analysis reveals no significant difference in the endolysosomal compartment between control (L) and *epg5* RNAi (M) nephrocytes. α: late endosomes; β: lysosomes; m: mitochondria; LD: lipid droplets.

observed Atg8a positive structures that colocalized with Epg5-9xHA, but not with Lamp1-3xmCherry, as well as structures that were triple positive (**Fig. 5K**), indicating that Epg5 is present on both autophagosomes and autolysosomes. Epg5 has been suggested as a Rab7 effector both in fly (Gillingham et al., 2014) and in mammalian cells (Wang et al., 2016), which we could confirm as Epg5-9xHA coprecipitates with Rab7-FLAG in cultured fly cells (**Fig. 5L**). Moreover, Epg5-9xHA also coprecipitates with the endogenous dynein motor Dhc64C (**Fig. 5M**), supporting the idea that Rab7 via Epg5 is required for dynein-dependent autophagosome transport. Taking into consideration that Epg5 was found to regulate the positioning of autophagosomes but not lysosomes, we utilized garland nephrocytes to study its effect on endolysosome maturation. Nephrocytes maintain a constant rate of endocytosis, making them ideal tools to study the endolysosomal system (Lőrincz et al., 2016). We have previously shown that inhibited late endosome to lysosome maturation leads to the enlargement of the late endosomal compartment (Boda et al., 2019; Hargitai et al., 2025; Lőrincz et al., 2019; Lőrincz et al., 2016; Lorincz et al., 2017). However, we found that neither the Rab7 or FYVE-GFP positive endosomal nor the Lamp1 positive lysosomal compartment changed upon the expression of *epg5* RNAi in nephrocytes (**Figure 5—figure supplement 1A-G**). This was strengthened by ultrastructural analyses, which showed no obvious changes in the morphology of the endolysosomal compartment in *epg5* RNAi nephrocytes (**Figure 5—figure supplement 1H, I**). These results suggest that in flies, Epg5 functions primarily in the autophagic pathway, independently from the endosomal system.

Knockdown of other small GTPases that play essential roles in the lysosomal system, such as Rab2 (Lorincz et al., 2017), Rab5 (Poteryaev et al., 2010), Rab14 (Mauvezin et al., 2016), or Arl8 (Boda et al., 2019), did not change the perinuclear pattern of mCherry-Atg8a positive autophagosomes in *vps16a* RNAi cells (**Figure 4—figure supplement 2A-D, G**). Notably, silencing the recycling endosomal Rab11 resulted in the scattering of mCherry-Atg8a puncta and the exceptionally strong accumulation of autophagosomes revealed by endogenous Atg8a staining (**Figure 4—figure supplement 2E-H**). Rab11 has been shown to be involved in autophagosome maturation in flies (Szatmári et al., 2014), but neither Rab11 interactors resulted in any change in the perinuclear autophagosome distribution (**Supplementary File 1, Supplementary Table 1**), suggesting that the effects of Rab11 may be indirect, compared to our Rab7 and Rab39 hits.

Next, we analyzed the effect and localization of overexpressed autophagosomal and endolysosomal Rabs in *vps16a* RNAi cells. Neither the wild type nor the constitutively active forms of the overexpressed Rabs changed the perinuclear distribution of mCherry-Atg8a positive autophagosomes (**Figure 4—figure supplement 3A, B, D-L, N**). Importantly, overexpression of both forms of YFP-tagged Rab7 (**Figure 4—figure supplement 3A, B**) and Rab2 (**Figure 4—figure supplement 3D, E**), as well as endogenous Rab7 immunolabeling (**Figure 4—figure supplement 3C**) showed obvious autophagosomal localization. Moreover, wild type YFP-Rab39 also overlapped with the mCherry-Atg8a puncta (**Figure 4—figure supplement 3F**) in *vps16a* RNAi cells. Our results suggest that Rab2 is exclusively required for autophagosomal fusions, while Rab7 and Rab39 are also required for autophagosome movement (**Figure 4—figure supplement 3M**). The YFP-tagged wild type form of Rab14, which was described as a regulator of autophagic vesicle transport and fusion (Mauvezin et al., 2016), exhibited a punctate pattern, but did not localize to autophagosomes, suggesting that it localizes to other organelles, most likely lysosomes (**Figure 4—figure supplement 3L**). Taken together, Rab7 and Rab39, as well as their effectors, Epg5 and ema, respectively, appear to be the most important regulators responsible for microtubular autophagosome motility in both directions.

Autophagosome transport machinery functions similarly in *snap29* and *vps16a* RNAi cells

While our findings strongly suggest that pre-fusion autophagosomes exhibit default minus-end-directed motility, we could not exclude the possibility that this phenotype specifically results from loss of the HOPS complex. To address this, we aimed to generate a *Drosophila* line with GFP-positive mosaic fat body cells expressing RNAi against one of the SNARE proteins required for autophagosome-lysosome fusion. Syntaxin17 (Syx17), the autophagosomal SNARE (Lőrincz & Juhász, 2020; Takáts et al., 2013), appeared to be the obvious choice. However, in Syx17 knockdown nephrocytes, we observed a tethering lock that permanently anchors lysosomes and autophagosomes together (Hargitai et al., 2025), making it impossible to study autophagosome motility independently of lysosomes. Therefore, we turned to the SNARE Snap29 as an alternative and generated a *Drosophila* line with GFP-positive mosaic fat body cells expressing Snap29 RNAi. Like our *vps16a* RNAi line used in the genetic screen, these flies also expressed an mCherry-Atg8a reporter driven by a UAS-independent, fat body-specific R4 promoter. Snap29 encodes a SNARE protein essential for autophagosome-lysosome fusion (Lőrincz & Juhász, 2020; Takáts et al., 2013).

We crossed this line with a control (luciferase) RNAi and with several key hits from our screen. Similar to *vps16a* RNAi, Snap29 knockdown led to perinuclear accumulation of autophagosomes (Fig. 6A, I). In *snap29* RNAi cells, *shot* and *dhc64c* RNAi redistributed autophagosomes to an ectopic MTOC or to the periphery, respectively (Fig. 6B, C, I). Knockdown of the kinesin Khc did not affect the perinuclear accumulation of autophagosomes in *snap29* RNAi cells (Fig. 6D, I), while RNAi targeting the small GTPases Rab7, Rab39, or their adaptors (Epg5 or ema) caused autophagosomes to become dispersed throughout the cytosol (Fig. 6E–I).

These results indicate that the redistribution of autophagosomes toward the microtubule minus-end is a general consequence of impaired fusion, and further suggest that the autophagosome transport machinery functions independently of Vps16A and the HOPS complex.

The transport of pre-fusion lysosomes is also minus-end directed

Given that *vps16a* RNAi led to the perinuclear distribution of immature lysosomes (Figure 1—figure supplement 1C) similar to autophagosomes, we hypothesized that pre-fusion autophagosomes and lysosomes travel in the same orientation, potentially sharing the same transport machinery. Thus, we stained lysosomes and co-expressed *vps16a* RNAi along with RNAi targeting our hits from the autophagosome positioning screen (Fig. 7). Importantly, in most cases, we observed similar phenotypes with Lamp1 staining as we did with Atg8a: spectraplaklin (*shot*) RNAi redistributed Lamp1 organelles to an ectopic ncMTOC, dynein inhibition redistributed lysosomes to the periphery, *rab7*, *rab39*, and *ema* RNAi-s resulted in the scattering of lysosomes across the cytosol, and kinesin depletion had no effect on the perinuclear accumulation of lysosomes (Fig. 7).

However, there were two important exceptions: Epg5 knockdown left lysosomes perinuclear (Fig. 7C, L), suggesting that it is indeed an autophagosomal adaptor. The other exception was *rab2* RNAi, which had no significant effect on autophagosome transport but resulted in dispersed, and sometimes even peripheral, lysosomal distribution in Vps16A-depleted cells (Fig. 7I, L). This result suggests that Rab2 is a potential regulator of minus-end directed transport of lysosomes and that pre-fusion organelles (autophagosomes and lysosomes) predominantly travel towards the minus-end of the MTs and share the main molecular components.

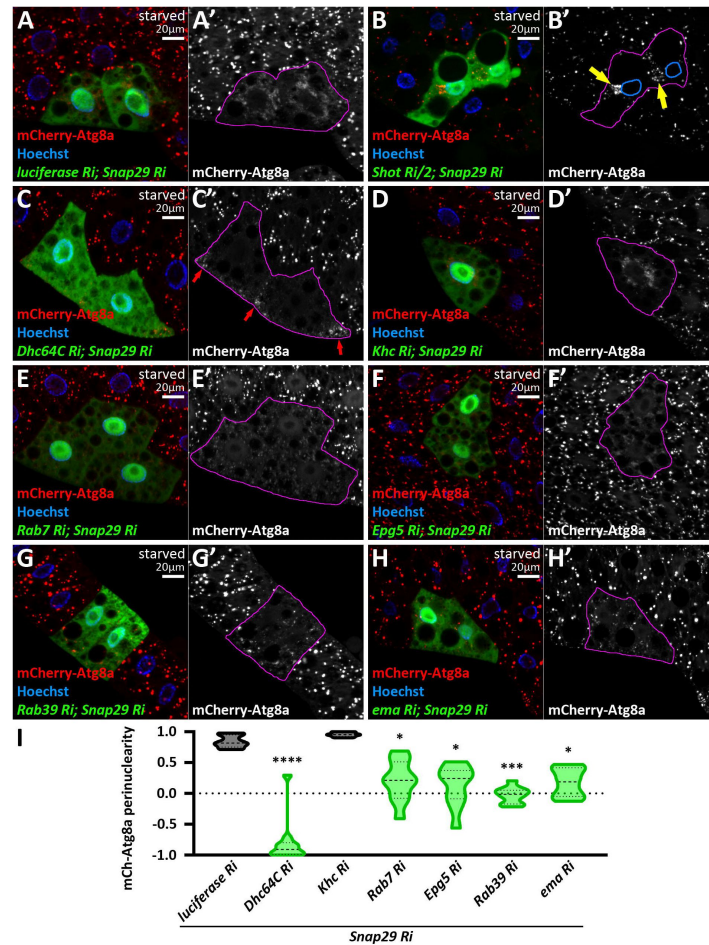


Figure 6.

A-H: Knockdown of key regulators of autophagosome transport in a Snap29 RNAi background results in autophagosome localization patterns similar to those observed with Vps16A RNAi.

A-H: In *luciferase*; Snap29 double knockdown cells, non-fused autophagosomes accumulate in the perinuclear region, marked by mCherry-Atg8a (A). *Shot*; Snap29 double knockdown causes autophagosomes to accumulate around an ectopic MTOC (B), marked by yellow arrows in B'. Nuclear outlines are shown in blue. *Dhc64C* knockdown in Snap29 RNAi cells causes autophagosomes to redistribute to the cell periphery (C, red arrows in C'). *Khc* knockdown does not alter the perinuclear distribution of autophagosomes seen in Snap29 RNAi cells (D). Co-knockdown of Snap29 with *Rab7* (E), *Epg5* (F), *Rab39* (G), or *ema* (H) results in scattered autophagosome distribution throughout the cytoplasm. In grayscale panels, the boundaries of RNAi-expressing cells are highlighted in magenta. **I:** Quantification of the data shown in panels A and C-H. n = 10 cells.

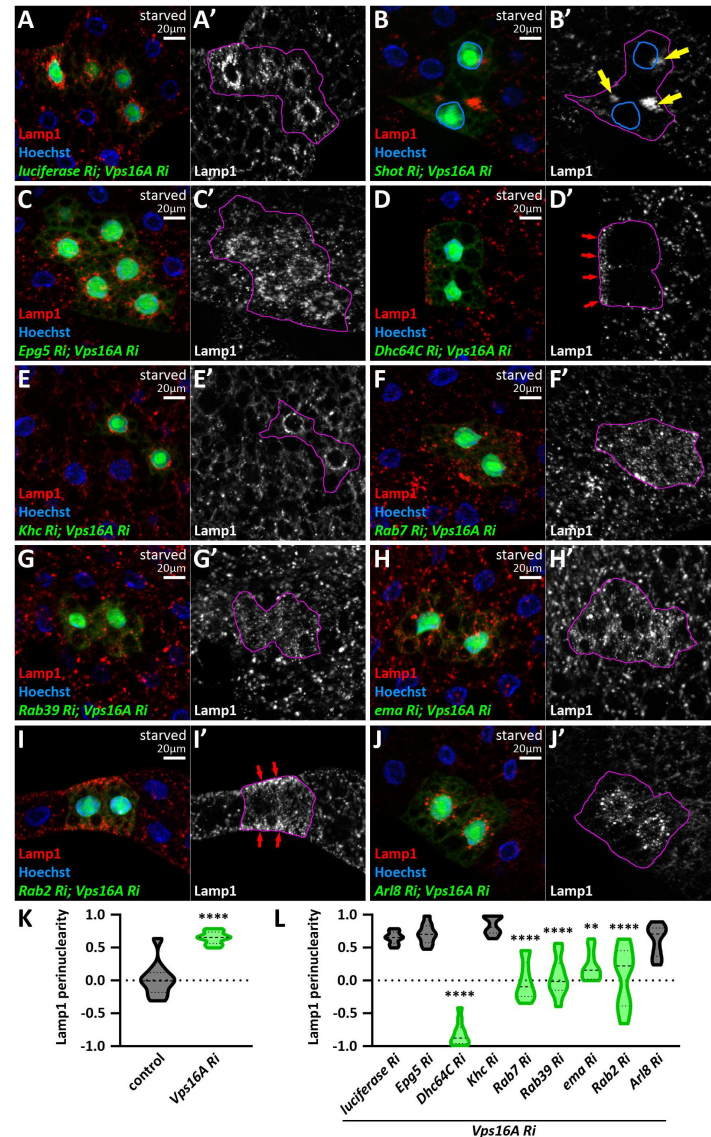


Figure 7.

The positioning of pre-fusion, immature autolysosomes is very similar to autophagosomes in *vps16a* RNAi cells.

A-J: Lamp1 positive lysosomes accumulate around the nuclei in cells co-expressing a control (*Luciferase*) RNAi (A). This positioning remains unaffected by the co-expression of *khc* (E) or *arl8* RNAi (J). Depletion of *Shot* (B) or *Dhc64C* (C) in *vps16a* RNAi cells redistributes Lamp1 positive lysosomes from the perinuclear cytoplasm to an ectopic MTOC (yellow arrows) or to the periphery, respectively. Lamp1 positive lysosomes are scattered throughout the cytosol in *vps16a* RNAi cells upon the co-expression of *rab7* (F), *rab39* (G), and *ema* (H) RNAi-s. In contrast, Lamp1 positive lysosomes retain their perinuclear distribution in *epg5*, *vps16a* double RNAi cells (D). Similar to *Rab7* or *Rab39*, the expression of *rab2* RNAi in *Vps16A* KD cells results in the scattering of Lamp1 positive lysosomes, with a trend observed that lysosomes tend to accumulate near the periphery (red arrows) (I). The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. The outlines of nuclei are drawn in blue in B'. **K, L:** Quantification of data shown in A, C-J; n=10 cells.

Minus-end directed autophagosome and lysosome transport is required for autophagosome-lysosome fusion

We hypothesized that concentrating pre-fusion autophagosomes and lysosomes at the perinuclear region increases the probability of their fusion, thereby promoting autolysosome formation. To test this, we used similar reporters and reagents as above but did not silence Vps16A in the examined cells to allow autophagosome-lysosome fusions. First, we analyzed Epg5 knockdown, which has been described to inhibit autolysosome maturation based on GFP-Atg8a (Byrne et al., 2016 [DOI](#)). In line with this, 3xmCherry-Atg8a positive autolysosomes were significantly smaller than in control cells, without any obvious change in their distribution (**Fig. 8A, B, E, F** [DOI](#)). Since 3xmCherry-Atg8a is transported to the lysosomes via autophagosome-lysosome fusion, and it retains its fluorescence in the lysosomal environment, it can be used to monitor autolysosome maturation. Although we cannot determine the number of lysosomes that fuse with each autophagosome, the overall size of 3xmCherry-Atg8a positive structures correlates with autolysosome maturation efficiency (Lőrincz et al., 2017 [DOI](#)). Accordingly, the large Rab7 and Arl8 positive autolysosomes were almost completely absent from Epg5 silenced cells, as revealed by immunostainings (**Fig. 8C, D, G, H** [DOI](#)), with only small autolysosomes present.

Silencing of the dynein and dynactin hits: Dhc64C, sw, Dlic, and robl, as well as DCTN1-p150, resulted in the redistribution of 3xmCherry-Atg8a positive autolysosomes to the periphery and their size became significantly smaller (**Fig. 9A-E, I, K** [DOI](#)), suggesting that loss of minus-end directed transport leads to autolysosome maturation defects. In contrast, *khc* and *klc* RNAi caused the perinuclear accumulation of autolysosomes, which appeared larger than those in the controls, suggesting a trend, although this difference did not reach statistical significance (**Fig. 9F, G, J, L** [DOI](#)).

In accordance with its suggested role in autophagosome-lysosome fusion, silencing of Rab7 resulted in significantly smaller and mostly dispersed autolysosomes (**Figure 9—figure supplement 1A, D, E** [DOI](#)) (Hegedus et al., 2016 [DOI](#); Lőrincz et al., 2017 [DOI](#)). Rab39 knockdown, however, led to mostly peripheral autolysosomes, which were not different in size from those in the control cells (**Figure 9—figure supplement 1B, D, E** [DOI](#)), but the localization of the mCherry-Atg8a positive autolysosomes suggests that Rab39 is also required for minus-end directed movement of mature lysosomes. Surprisingly, loss of *ema* did not influence the size and distribution of autolysosomes (**Figure 9—figure supplement 1C-E** [DOI](#)).

Our most important finding came when we silenced the spectraplakin Shot, which caused the accumulation of autolysosomes in the ectopic cytosolic MTOC (**Fig. 9H** [DOI](#)). Notably, their size significantly increased (**Fig. 9H, L** [DOI](#)). This can be explained by the fact that the volume surrounding the ectopic ncMTOC in Shot-depleted cells is smaller than the volume around the nuclei, leading to an increased fusion rate in these cells. Taken together, these results demonstrate that minus-end transport is crucial for proper autolysosome maturation.

The observation that dispersing autophagosomes and lysosomes under the plasma membrane in dynein/dynactin-silenced cells leads to insufficient autolysosome maturation, while concentrating autophagosomes and lysosomes to an ectopic ncMTOC in *shot* RNAi results in the enlargement of lysosomes, indicates that the autophagosome-lysosome fusion rate depends on the volume of cytoplasm in which these organelles meet. Several studies have suggested a connection between microtubular transport and the fusion of autophagosomes (Fass et al., 2006 [DOI](#); Jahreiss et al., 2008 [DOI](#); Kimura et al., 2008 [DOI](#); Köchl et al., 2006 [DOI](#)).

We tested this hypothesis by analyzing the colocalization between the 3xmCherry-Atg8a reporter and the lysosomal membrane protein Lamp1, either by immunostaining or by expressing GFP-Lamp1. Their overlap represents autolysosomes, while 3xmCherry-Atg8a positive, Lamp1 negative structures are considered pre-fusion autophagosomes. Conversely, Lamp1 positive, 3xmCherry-

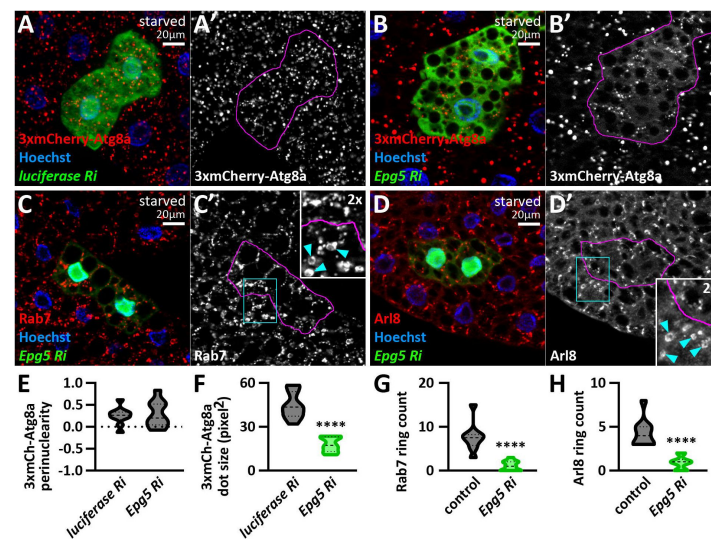


Figure 8.

Epg5 regulates autolysosome maturation.

A, B: Epg5 knockdown results in a significant reduction in the size of 3xmCherry-Atg8a positive autolysosomes (B) compared to control RNAi (*luciferase* RNAi) expressing cells (A). **C, D:** *epg5* RNAi cells lack large Rab7 (C) and Arl8 (D) positive autolysosomes, which are present in surrounding control cells (cyan arrowheads in insets point to Rab7 and Arl8 positive autolysosomes in control cells). The boundaries of RNAi cells are highlighted in magenta in the grayscale panels. **E-H:** Quantification of data shown in A-D; n=10 cells.

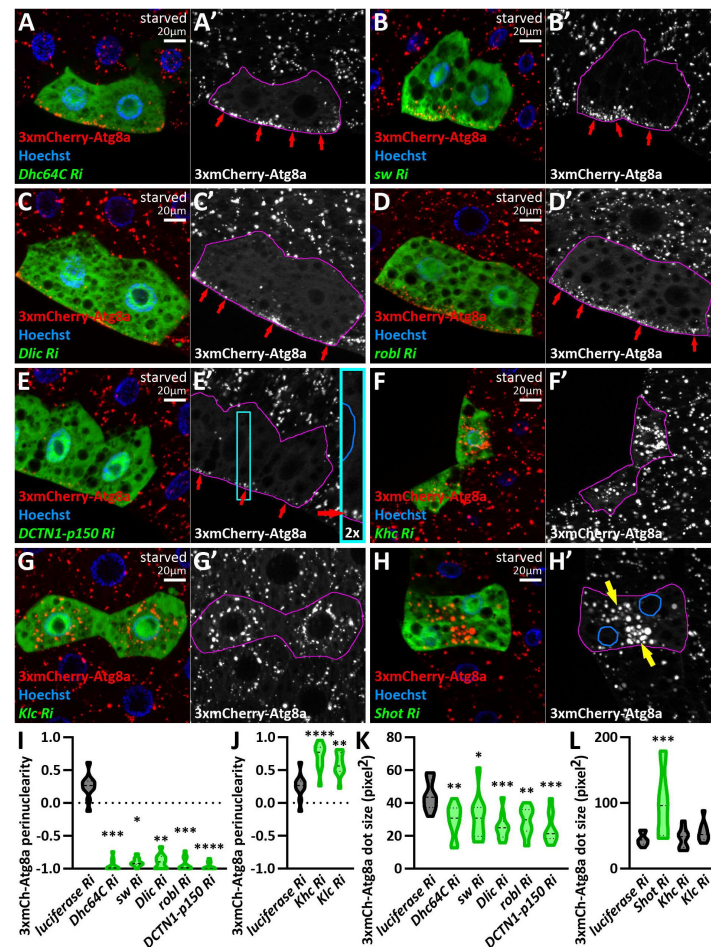


Figure 9.

Minus-end directed transport is required for autolysosome maturation.

A-E: Autolysosome size is significantly reduced upon the loss of dynein (A-D) or dynactin (E) function. Red arrows point to autolysosomes at the cell periphery in A'-E' and in the inset of E. **F, G:** Kinesin knockdowns do not significantly influence autolysosome size. **H:** Autolysosome size increases at ectopic foci (yellow arrows) in *shot* RNAi cells. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. The outlines of nuclei are drawn in blue in the inset of E' and in H'. **I-L:** Quantification of data shown in A-H; n=10 cells.

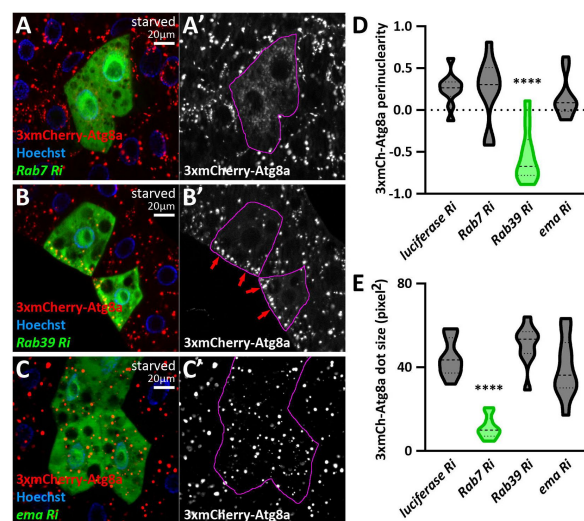


Figure 9—figure supplement 1.

A: Silencing of Rab7 decreases mCherry-Atg8a positive autolysosome size as expected, and they remain generally dispersed. **B, C:** Knockdown of Rab39 (B), but not ema (C), relocates mCherry-Atg8a positive autolysosomes to the periphery (red arrows). Importantly, neither knockdown decreases the size of autolysosomes. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. **D, E:** Quantification of data shown in A-C; n=10 cells.

Atg8a negative structures indicate lysosomes of non-autophagic origin. Thus, reduced colocalization indicates a fusion defect. In control cells, Lamp1 or GFP-Lamp1 overlaps with mCherry-Atg8a, indicating that these organelles are indeed autolysosomes (Fig. 10A, H [↗](#), Figure 10—figure supplement 1A, G [↗](#)). In *Shot*-depleted cells, both signals overlapped in the ectopic ncMTOC, indicating that autophagosomes can effectively fuse with lysosomes in this region (Fig. 10B, H [↗](#), Figure 10—figure supplement 1B, G [↗](#)). Accordingly, the loss of the kinesin heavy chain *Khc* did not alter the overlap of these markers, proving that autolysosomes could still be formed (Fig. 10C, H [↗](#), Figure 10—figure supplement 1C, G [↗](#)). Importantly, the loss of the dynein motor *Dhc64C*, as well as the dynactin subunit *DCTN1-p150*, greatly reduced the overlap of signals, indicating a less effective autophagosome-lysosome fusion (Fig. 10D, E, H [↗](#), Figure 10—figure supplement 1D, G [↗](#)). Since *Rab7* and *Epg5* have been implicated in autolysosome maturation, their depletion reduced the overlap of autophagic and lysosomal markers (Byrne et al., 2016 [↗](#); Hegedus et al., 2016 [↗](#); Wang et al., 2016 [↗](#)) (Fig. 10F-H [↗](#), Figure 10—figure supplement 1E-G [↗](#)). Although loss of minus end directed transport significantly impaired autophagosome-lysosome fusion, the degree of inhibition did not reach the level observed upon *Vps16A* knockdown (Figure 10—figure supplement 1H-K [↗](#)). This indicates that impaired motility alone does not directly block fusion, but rather reduces its probability. Taken together, our results suggest that the role of minus-end directed transport in fat cells is to bring pre-fusion organelles into proximity to increase the likelihood of their fusion.

Discussion

Microtubular transport of different organelles within the lysosomal system is essential for their proper function. Among the small GTPases that regulate late endosome/lysosome positioning, *Rab7* (Fujiwara et al., 2016 [↗](#); Jordens et al., 2001 [↗](#); Ma et al., 2018 [↗](#); van der Kant et al., 2013 [↗](#)) and *Arl8* (Bagshaw et al., 2006 [↗](#); Boda et al., 2019 [↗](#); Hofmann & Munro, 2006 [↗](#); Marwaha et al., 2017 [↗](#); Rosa-Ferreira & Munro, 2011 [↗](#); Rosa-Ferreira et al., 2018 [↗](#)) are well-studied. However, the positioning of autophagosomes is less understood. Autophagosome transport has been primarily studied in highly polarized neurons under basal conditions. In neuronal cells, dynein-mediated transport, regulated by several suggested adaptor molecules, has been described. However, most reporters and reagents used for studying autophagosome transport cannot differentiate between pre-fusion and post-fusion autophagic vesicles. Therefore, the regulation of non-fused autophagosome motility remains unclear. We propose that an autophagosome should not be considered as such once its lumen has started to acidify or undergo fusion, transitioning to a maturing autolysosome.

To address this issue, we established a genetic system generating mosaic cells where *Vps16A*, a central HOPS subunit, is silenced. Without *Vps16A*, cells are unable to fuse autophagosomes with lysosomes or late endosomes (Takáts et al., 2014 [↗](#)), thus preventing the formation of autolysosomes. This system allowed us to study autophagosome positioning without misidentifying autolysosomes as autophagosomes. Given that fat cells have perinuclear non-centrosomal MTOCs, we hypothesized that non-fused autophagosomes move in a minus-end direction, likely driven by dyneins. This hypothesis was supported by our observation that relocating the ncMTOC using *shot* RNAi also relocated autophagosomes to this region.

Loss of dyneins and dynactins not only blocked perinuclear autophagosome positioning but also redistributed them to the cell periphery, suggesting that kinesin-regulated motility becomes available for starvation-induced autophagosomes in these conditions. Importantly, we showed that not all dyneins or dynactins are required for this transport. A dedicated cytosolic dynein complex, composed of *Dhc64* (a heavy chain subunit), *sw* (an intermediate chain subunit), *Dlic* (a light intermediate chain subunit), and *robl* (a light chain subunit), activated by a dynactin complex, is required for autophagosome transport. This is consistent with observations that autophagosomes can move bidirectionally in neurons and that purified autophagosome fractions contain both

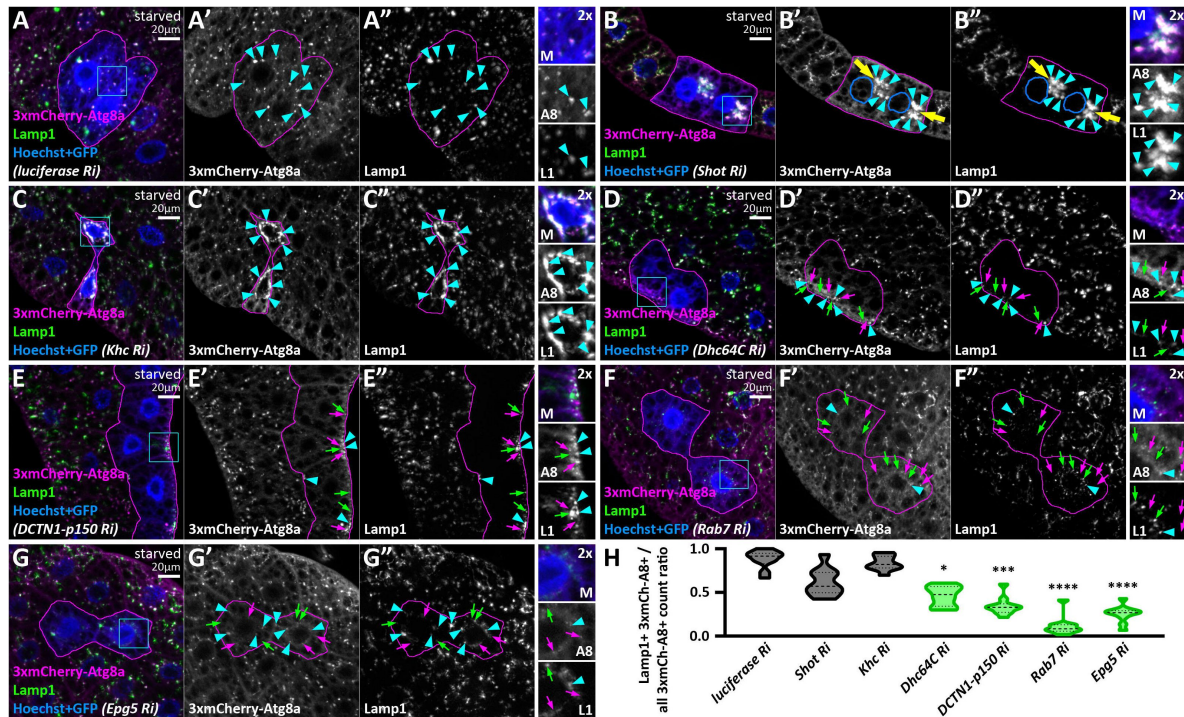


Figure 10.

Loss of the autophagosome positioning machinery decreases autophagosome-lysosome fusion.

A-G: In starved control RNAi (*luciferase*) expressing cells, mCherry-Atg8a overlaps with endogenous Lamp1 (A), indicating normal autophagosome-lysosome fusion and autolysosome formation. Autolysosomes still form in *shot* (B) or *khc* RNAi (C) cells, as mCherry-Atg8a colocalizes with endogenous Lamp1 similar to controls, but these are found in ectopic foci (yellow arrows) in *shot* RNAi cells. The outlines of nuclei are drawn in blue in B' and B''. Conversely, RNAi-s targeting factors responsible for minus-end directed autophagosome transport (D-G) decrease this overlap, suggesting less effective autophagosome-lysosome fusion. The GFP signal of RNAi-expressing cells is false-colored blue in composite images. The boundaries of RNAi-expressing cells are highlighted in magenta. The boxed areas in the main panels, marked by cyan, are enlarged in the insets (M, merged image; A8, 3xmCherry-Atg8a; L1, Lamp1). Cyan arrowheads point to mCherry-Atg8a/Lamp1 double-positive structures, while magenta and green arrows indicate mCherry-Atg8a or Lamp1 single-positive dots, respectively. **H:** Quantification of data shown in A-G; n=10 cells.

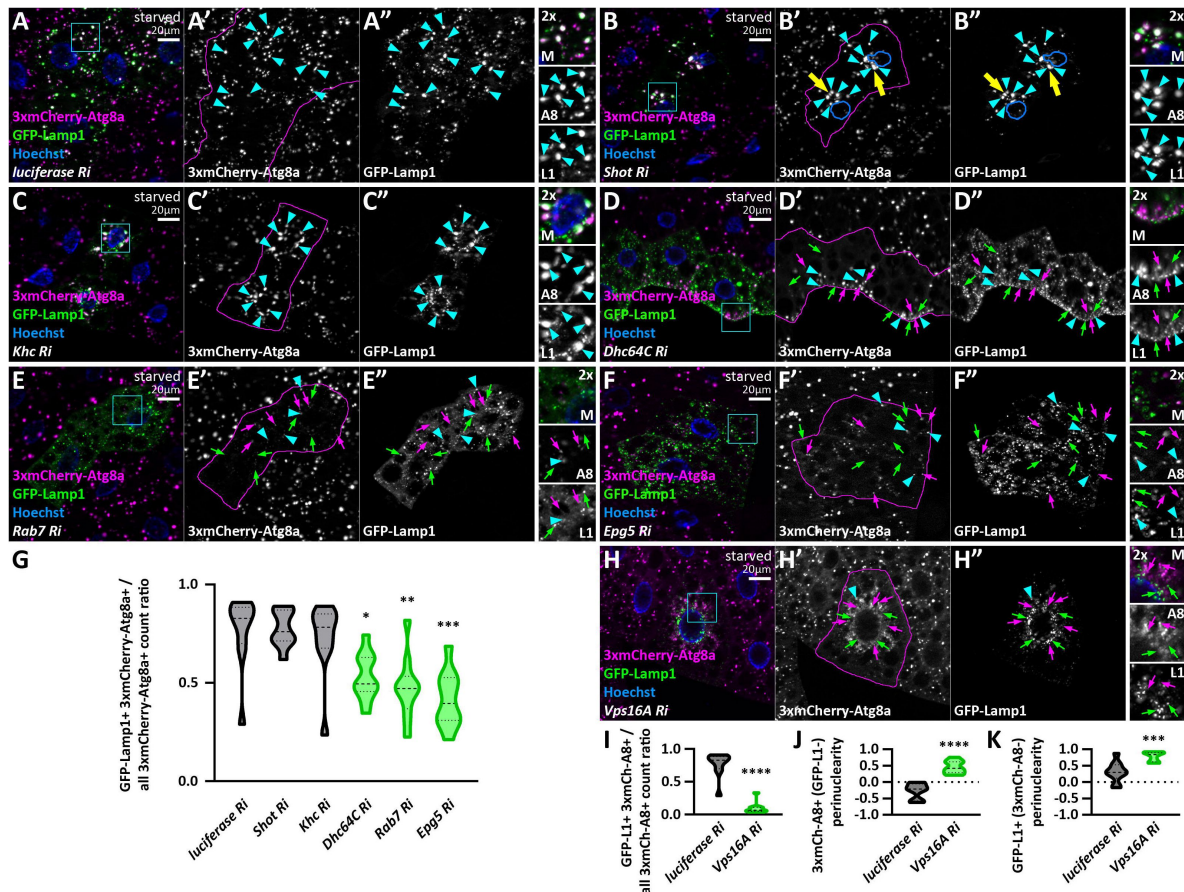


Figure 10—figure supplement 1.

A-F: Partial inhibition of autophagosome-lysosome fusion is observed upon knockdown of factors responsible for minus-end transport, based on 3xmCherry-Atg8a and GFP-Lamp1 colocalization experiments. In starved control RNAi (*Luciferase*) expressing cells, mCherry-Atg8a overlaps with GFP-Lamp1 (A), indicating normal autophagosome-lysosome fusion and autolysosome formation. Autolysosomes still form in *shot* (B) or *khc* RNAi cells (C), as mCherry-Atg8a colocalizes with GFP-Lamp1, but these are found in ectopic foci (yellow arrows) in the former case. The outlines of nuclei are drawn in blue in B', B''. Conversely, RNAi-s targeting factors responsible for minus-end directed autophagosome transport (D-F), including *epg5* RNAi, decrease this overlap, suggesting less effective autophagosome-lysosome fusion. **G:** Quantification of data shown in A-F; n=10 cells. **H:** Silencing of *Vps16A*, used as a positive control, almost completely blocks autophagosome-lysosome fusion as indicated by the minimal overlap of the two signals. **I-K:** Quantification of data shown in H; n=10 cells. The boundaries of RNAi-expressing cells are highlighted in magenta in the grayscale panels. The boxed areas in the main panels, marked by cyan, are enlarged in the insets (M, merged image; A8, 3xmCherry-Atg8a; L1, GFP-Lamp1). Cyan arrowheads show overlapping dots, while magenta and green arrows point to non-colocalizing 3xmCherry-Atg8a positive and GFP-Lamp1 positive dots, respectively.

dyneins and kinesins (Maday et al., 2012). Our results also reinforce findings that microtubule inhibitors block centrosome-directed autophagosome transport in mammalian cells (Fass et al., 2006).

Taking these observations into consideration, we investigated the relationship between dyneins and kinesins in autophagosome transport. Overexpression of kinesin motors blocked minus-end transport, while autophagosomes in kinesin and dynactin double knockdown cells appeared immobile. Moreover, expressing a recombinant minus-end kinesin (Clark et al., 1997) partially rescued the peripheral relocation of autophagosomes upon dynactin silencing, indicating a competitive relationship between minus- and plus-end motors in autophagosome positioning. The possibility of plus-end transport as a secondary mechanism raises questions about its physiological role. Besides enabling bidirectional movement (Jahreiss et al., 2008; Maday et al., 2012), it is possible that autophagosomes transport kinesins to autolysosomes, similar to endosomes, which are suggested to transport dyneins to autophagic vacuoles (Cheng et al., 2015). Therefore, if kinesins are present but are downregulated on autophagosomes, the absence of dyneins could potentially release them from inhibition. Autophagic vesicles are suggested to move towards the plus-end (Mauvezin et al., 2016; Pankiv et al., 2010), and loss of various kinesins leads to autophagosome accumulation in the cell center in mammalian cells (Cardoso et al., 2009; Korolchuk et al., 2011). The plus-end transport of autophagic vesicles by the Klp98A kinesin in *Drosophila* promotes autophagosome-lysosome fusion and degradation (Mauvezin et al., 2016). However, our tests indicated that Rab14 and its interactor Klp98A likely transport autolysosomes, not autophagosomes, as evidenced by the non-autophagosomal localization of YFP-Rab14 and that their RNAi-s had no effect on autophagosome positioning in starved, HOPS-depleted cells. Since neither kinesin RNAi altered the perinuclear accumulation of autophagosomes in Vps16A-depleted cells, we propose that the default direction is towards the MTOC at the minus-ends of microtubules.

Among Rab small GTPases, we identified Rab7 and Rab39 as regulators of autophagosome positioning. Rab7, crucial for autophagosome-lysosome fusion, appears on autophagosomes (Hegedus et al., 2016). Its knockdown resulted in autophagosomes remaining randomly positioned in the cytosol, highlighting its importance in bidirectional motility. Consistent with our findings, Rab7 regulates both minus- and plus-end directed motility of lysosomes or endosomes, involving several adaptors such as Plekhm1 and FYCO1 (Fujiwara et al., 2016; Jordens et al., 2001; Ma et al., 2018; Pankiv et al., 2010; Tabata et al., 2010; van der Kant et al., 2013). However, only one Rab7 interactor, the autophagy adaptor Epg5 (Gillingham et al., 2014; Wang et al., 2016), produced a similar phenotype to Rab7 in our tests. Additionally, we demonstrated that Epg5 coprecipitates with Dhc64C, and localizes to Rab7 and Atg8a-positive vesicles in cultured fruit fly cells. As Epg5 loss did not significantly impact the endolysosomal system, it appears to function in the autophagic pathway. Epg5 interacts with Rab7 and LC3 to mediate autophagosome-lysosome fusion in fly, worm and mammalian cells (Hori et al., 2017; Wang et al., 2016), and its mutations are linked to Vici syndrome, a severe neurodegenerative disorder in humans (Balasubramaniam et al., 2018; Byrne et al., 2016; Meneghetti et al., 2019). We thus identify a potential new role for Epg5 in autophagosome positioning.

Silencing Rab39 and its interactor ema (Gillingham et al., 2014) produced a phenotype similar to Rab7 or Epg5 loss. Rab39 regulates lysosomal function and interacts with HOPS in mammalian cells (Lakatos et al., 2021; Zhang et al., 2023). Ema promotes autophagosome biogenesis (Kim et al., 2012) and endosomal maturation through HOPS interaction (Kim et al., 2010). Our results suggest that Rab7-Epg5 and Rab39-ema interactions are both necessary for bidirectional autophagosome motility. Further studies are required to clarify their exact roles and interrelations. YFP-tagged Rab7 and Rab39 both colocalized with autophagosomes, supporting their role in motility. Rab2, Rab7, and Arl8 are known lysosomal fusion factors (Boda et al., 2019; Hegedus et al., 2016; Lorincz et al., 2017), but unlike Rab7, neither Rab2 nor Arl8 knockdown affected autophagosome positioning. Since YFP-Rab2 was also found to colocalize with

autophagosomes, our results suggest that its sole role on autophagosomes is to regulate maturation. Although our RNAi screen identified a molecular machinery potentially sufficient for autophagosome transport, other small GTPases, adaptors, or motor subunits may also influence this process. Because validating the proper loss-of-function effect for every RNAi line used in the screen was impractical, we cannot exclude the possibility that some negative results were due to inefficient silencing. It is also important to note that *Drosophila* cells exhibit a wide variety of microtubule-organizing centers (MTOCs) during development, including both centrosomal and non-centrosomal types (Tillery et al., 2018 [DOI](#)). A key future direction will be to examine autophagosome motility in additional cell types to determine whether the same positioning machinery operates universally, or whether cell type-specific mechanisms exist.

Given the small Arl8-positive lysosomes in the perinuclear region upon Vps16A silencing, we hypothesized that pre-fusion lysosomes might exhibit similar motility to autophagosomes. This was largely confirmed, as minus-end transport of immature lysosomes depended on dyneins, Rab7, Rab39, and ema. However, Epg5 did not influence lysosome positioning, indicating that its function is likely exerted at the surface of autophagosomes. Conversely, Rab2 was identified as a regulator of minus-end directed lysosome transport. These findings suggest similar but distinct regulatory mechanisms for pre-fusion organelle transport, possibly to enhance fusion efficiency by converging them towards the cell center. Rab2's interaction with motor adaptors such as Bicaudal D (Gillingham et al., 2014 [DOI](#)) likely regulates lysosome motility.

Moreover, mature autolysosomes, but not pre-fusion ones, redistributed to the cell periphery upon Rab39 silencing, similar to dynein-depleted cells, suggesting that Rab39 exclusively regulates minus-end movement at the post-fusion level. Considering that plus-end directed lysosome motility is regulated by Arl8 (Bagshaw et al., 2006 [DOI](#); Boda et al., 2019 [DOI](#); Hofmann & Munro, 2006 [DOI](#); Marwaha et al., 2017 [DOI](#); Rosa-Ferreira & Munro, 2011 [DOI](#); Rosa-Ferreira et al., 2018 [DOI](#)), which is dispensable for autophagosome motility, it is plausible that Rab39 promotes bidirectional transport of pre-fusion organelles, while its role post-fusion is restricted to minus-end directed motility.

An important question is why pre-fusion autophagosomes and lysosomes travel to the same destination. It is feasible to think this is because they need to fuse with each other. We found that loss of minus-end directed motility reduced autolysosome maturation; in dynein- or dynactin-depleted cells, only smaller autolysosomes were produced, and autophagosome-lysosome fusion decreased. This aligns with previous observations that dynein-regulated autophagosomal motility is indispensable for efficient lysosomal fusion (Jahreiss et al., 2008 [DOI](#); Kimura et al., 2008 [DOI](#)). Conversely, inhibiting kinesins did not impede autophagosome-lysosome fusion, and concentrating autophagosomes and lysosomes to a smaller ectopic ncMTOC via *shot* RNAi resulted in enlarged lysosomes, allowing autophagosome-lysosome fusion to proceed.

Therefore, we propose a model in which pre-fusion organelles travel towards the MTOC in a cytosolic dynein-dependent manner regulated by small GTPases and their adaptors (Rab7-Epg5 and Rab39-ema on autophagosomes; Rab7, Rab39-ema, and Rab2 on lysosomes) to enhance fusion probability. After fusion, autolysosomes can move to the periphery in an Arl8-dependent manner and back, regulated by Rab39 (Fig. 11 [DOI](#)).

Materials and methods

Fly work and RNAi based screen

We raised the fly stocks and crosses in glass vials, on standard food at 25°C. Early third instar larvae were starved for 3 hours in 20% sucrose solution. Next fat bodies were dissected in cold PBS, mounted in a 8:2 mixture of glycerol and PBS completed with Hoechst 33342 as nuclear dye (5

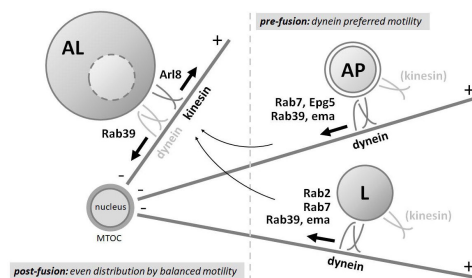


Figure 11.

Model of the transport of autophagosomes and lysosomes in starved fat cells.

Before fusion, autophagosomes and lysosomes are transported towards the perinuclear ncMTOC by a cytosolic dynein complex in starved fat cells to ensure proper fusion and effective degradation. This process requires Rab7, Rab39, and their interactors Epg5 and ema on autophagosomes, and Rab2, Rab7, Rab39, and the Rab39 interactor ema, but not Epg5, on lysosomes. After fusion, Arl8 mediates the plus-end transport of autolysosomes, while Rab39 promotes dynein-regulated minus-end directed transport. Thus, the motility of pre-fusion and post-fusion organelles is differently regulated: pre-fusion organelles generally move towards the MTOC, while post-fusion organelles exhibit bidirectional motility. This spatial regulation ensures proper fusion rates and degradation efficiency.

µg/ml) (Thermo), then imaged immediately.

For the RNAi based genetic screen, we established the *hs-Flp; Vps16A RNAi, UAS-DCR2; act<CD2<Gal4, UAS-GFPnls, r4-mCherry-Atg8a* stock, in order to generate *vps16a* RNAi expressing fat cells. This was crossed with RNAi or overexpression lines of interest.

All the screened *Drosophila* lines, as well as their sources, identifiers and phenotypes are listed in **Supplementary Table 1**. Representative images of the phenotypes of screened lines are shown in **Supplementary File 1**. The proper genotypes and the fly stocks from the screen that were used in the Figure panels are summarized in **Supplementary Table 2**.

For further experiments, we used the following mosaic cell generating stocks, with or without *Vps16A* RNAi.:

- *hs-Flp; Vps16A RNAi, UAS-DCR2; act<CD2<Gal4, UAS-GFPnls,*
- *hs-Flp; UAS-DCR2; act<CD2<Gal4, UAS-GFPnls,*
- *hs-Flp; 3xmCherry-Atg8a, UAS-2xEGFP; act<CD2<Gal4, UAS-DCR2,*
- *hs-Flp; 3xmCherry-Atg8a, UAS-GFP-Lamp1; act<CD2<Gal4, UAS-DCR2.*

All of these stocks were described before (Boda et al., 2019; Lorincz et al., 2017; Takáts et al., 2013). The *prospero-Gal4* driver (80572; FlyBase ID: FBst0080572) and the *UAS-GFP-myc-2xFYVE* reporter (42712; FlyBase ID: FBst0042712) used for the garland cell experiments were obtained from Bloomington *Drosophila* Stock Center.

Immunohistochemistry

For immunohistochemistry experiments, we dissected and fixed the samples in 4% paraformaldehyde (in PBS) for 45 min, washed them in PBS for 2×15 min, permeabilized in PBTX (0.1% Triton X-100 in PBS) for 20 min and blocked in 5% fetal bovine serum (in PBTX) for 30 min. The samples were then incubated with the primary antibodies (diluted in the blocking solution) overnight at 4°C, followed by washing in PBTX containing 4% NaCl for 15 min, washing in PBTX for 2×15 min, blocking in 5% fetal bovine serum (in PBTX) for 30 min and incubating with the secondary antibodies for 3 hours. The samples were then washed in PBTX containing 4% NaCl and 5 µg/ml Hoechst 33342 for 15 min, in PBTX for 2×15min, and in PBS for 2×15 min. All steps, except the incubation with primary antibodies, were performed at room temperature.

In case of garland immunohistochemistry and Lamp1 immunostainings of fat bodies for the 3xmCherry-Atg8a colocalization experiment, samples were dissected in a buffer containing 80 mM PIPES, 5 mM EGTA and 1 mM MgCl₂ (pH was adjusted to 6.8 with NaOH) and fixed in this solution containing also 3.7% formaldehyde, 0.25% glutaraldehyde and 0.2% Triton X-100, for 45 min. Following the fixation, the samples were incubated with 2 mg/ml sodium borohydride (in PBS) for 2.5 min, washed in PBS for 2×15 min (once for 10 min in case of garland cells) and permeabilized in PBTX containing ammonium chloride and glycine (both 50 mM) for 20 min. The remaining steps were the same as described above.

For immunostaining S2R⁺ hemocytes (*Drosophila*), cells were fixed in 4% paraformaldehyde for 20 min, washed in PBS for 15 min, permeabilized in PBTX for 10 min, and blocked in 5% fetal bovine serum (in PBTX) for 30 min. The samples were then incubated with the primary antibodies overnight at 4°C, followed by washing in PBTX containing 4% NaCl for 15 min and in PBTX for 2×10 min, and incubation with the secondary antibodies (solved in the blocking solution) for 3 h. The cells were then washed in PBTX containing 4% NaCl and 5 µg/ml Hoechst 33342 for 15 min, in PBTX for 10 min and in PBS for 15 min. In case of Atg8a immunolabeling, cells were starved in a solution containing 10 mM D(+) glucose, 0.5 mM MgCl₂, 4.5 mM KCl, 121 mM NaCl, 0.7 mM Na₂HPO₄, 1.5 mM NaH₂PO₄, and 15 mM NaHCO₃ (pH 7.4) (Aguilera-Gomez et al., 2017).

The following primary antibodies were used: rat anti-Atg8a (1:800 or in case of S2R+ cells, 1:300; (Takáts et al., 2013 [DOI](#))); rabbit anti- β -galactosidase (1:100; Merck); goat anti-Gmap (1:1000; Developmental Studies Hybridoma Bank [DSHB]); mouse anti-Rab7 (1:10; DSHB; (Riedel et al., 2016 [DOI](#))); rabbit anti-Arl8 (1:300; DSHB); rabbit anti-Lamp1 (1:1000; (Chaudhry et al., 2022 [DOI](#))), rat anti-mCherry (1:300; (Takáts et al., 2014 [DOI](#))); guinea pig anti-mCherry (1:500); rabbit anti-HA (1:200; Merck); rabbit anti-HA (1:200; Proteintech) and chicken anti-GFP (1:1500; Invitrogen).

We used the following secondary antibodies: Alexa Fluor 568 goat anti-rat (1:1000); Alexa Fluor 647 donkey anti-rabbit (1:600); Alexa Fluor 568 donkey anti-goat (1:1000); Alexa Fluor 568 donkey anti-mouse (1:1000); Alexa Fluor 647 donkey anti-mouse (1:600); Alexa Fluor 568 donkey anti-rabbit (1:1000); Alexa Fluor 488 goat anti-chicken (1:1000); Alexa Fluor 488 donkey anti-rat (1:1000); Alexa Fluor 488 goat anti-rabbit (1:1000) (all Invitrogen) and DyLight 550 goat anti-guinea pig (1:600; Thermo Fisher).

Electron microscopy

For correlative ultrastructural analysis, fat bodies were dissected in a fixative containing 3.2% paraformaldehyde, 1% glutaraldehyde, 1% sucrose and 0.003 M CaCl_2 (in 0.1 N sodium cacodylate buffer, pH 7.4) on poly-L-lysine coated glass slides and fluorescent images were taken to help recognizing GFP and RNAi expressing cells. Then the samples were fixed in the same solution overnight at 4°C, then were post-fixed in 0.5% osmium tetroxide for 1 h, followed by half-saturated aqueous uranyl acetate for 30 min and dehydrated in a graded series of ethanol, followed by embedding into Durcupan ACM (Sigma-Aldrich) on the glass slides. RNAi cells in the embedded samples were identified in semi-thick sections stained with toluidine blue, then ultra-thin sections of 70 nm were cut, then stained with Reynold's lead citrate (8 min, RT).

Preparation of samples for ultrastructural analysis of garland nephrocytes was performed as described before (Lőrincz et al., 2016 [DOI](#)).

Images were taken by a JEOL JEM-1011 transmission electron microscope operating at 80kV, equipped with a Morada camera (Olympus) and iTEM software (Olympus).

Molecular cloning and biochemistry

Cloning

To generate *genEpg5-9xHA*, genomic DNA from *w1118 Drosophila* strain was isolated and used as a template. The genomic region containing *Drosophila CG14299* was amplified using primers 5'-CCAAGCTTGCATGCGGCCGATTTTCTGTGCGCGACTGTTG-3' and 5'-TAAAAGATGCGGCCGTTACCGCCTCCACCCGTGGCCATTAAGTGAATTC-3' and cloned into *pGen-9xHA* (Lőrincz et al., 2016 [DOI](#)) as a NotI-Acc65I fragment by using Gibson Assembly kit (New England Biolabs, Ipswich, MA).

To obtain N-terminally 3xFLAG-tagged Rab7, the coding region of Rab7 was amplified from *Drosophila* cDNA (GH03685 (DGRC Stock 7144; <https://dgrc.bio.indiana.edu/stock/7144> [DOI](#); RRID:DGRC_7144)) using primers 5'-ACAAGCGGCCGCGAGGTATGTCCGACGTAAGAAATCC-3' and 5'-TCTAGAGGTACCTTAGCACTGACAGTTGTCAGGA-3' and cloned into NotI-Acc65I sites of *pUAST-3xFLAG* vector (Takáts et al., 2014 [DOI](#)).

S2R+ maintenance and transfection

The S2R+ *Drosophila* cell line (*Drosophila* Genomics Resource Center; Stock 150; RRID: CVCL_Z831) was maintained in Insect XPress medium (Lonza) containing 10% FBS (EuroClone) and 1% Penicillin-Streptomycin (Lonza) at 26°C. Cells were transfected with *genEpg5-9xHA* plasmid using

Calcium phosphate method. DNA was diluted in 240 mM CaCl_2 , mixed with 2x HEPES-buffered saline (50 mM HEPES, 1.5 mM Na_2HPO_4 , 280 mM NaCl, pH 7.1), incubated at 25°C for 30 minutes, and added to the cells. Co-transfection with *genEpg5-9xHA* and *genLamp1-3xmCherry* (Hegedus et al., 2016) was performed using jetOPTIMUS® DNA Transfection Reagent (Polyplus). 24 hours after transfection, cells were used for immunohistochemistry or immunoprecipitation. In experiments, when cells were transfected with *pUAST-3xFLAG-Rab7* and *pGen-Epg5-9xHA* constructs, metallothionein-Gal4 plasmid was also applied. Protein expression was induced 24h after transfection with 500 μM CuSO_4 for overnight incubation.

Immunoprecipitation

Cells were transfected with appropriate plasmid constructs and were collected 24 hours after transfection. They were washed with PBS and lysed on ice in lysis buffer (0.5% Triton X-100, 150 mM NaCl, 5 mM EDTA, and 50 mM Tris-HCl, pH 7.5, complete protease inhibitor cocktail (Roche)) for 20 minutes. Cell lysates were cleared by centrifugation for 10 min at 20,000 g, 4°C, followed by the addition of mouse anti-HA or anti-FLAG agarose (Sigma-Aldrich) to the supernatant. After incubation at 4°C for 2 h, beads were collected by centrifugation at 5,000 g for 2 minutes at 4°C, followed by extensive washes in wash buffer (lysis buffer without protease inhibitors) and finally boiling in Laemmli sample buffer. Samples were analyzed by Western blot using rat anti-HA (1:1000; Roche), mouse anti-FLAG (M2; 1:2000; Sigma-Aldrich) and mouse anti-Dhc (2C11-2; 1:12.5; DSHB) antibodies. It is experimentally demonstrated that the Dhc antibody recognizes a polypeptide at around 260 kDa (Baker et al., 2021).

Imaging, quantification and statistics

We obtained the fluorescent images with an AxioImager M2 microscope (Zeiss), equipped with an ApoTome2 grid confocal unit (Zeiss) and with an Orca Flash 4.0 LT sCMOS camera (Hamamatsu), using Plan-Apochromat 40 $\times$ /0.95 NA Air and Plan-Apochromat 63 $\times$ /1.40 NA Oil objectives (Zeiss), and Zeiss Efficient Navigation 2 software. Images from eleven consecutive focal planes (section thickness: 0.35 μm in case of the 40 $\times$ objective, and 0.25 μm in case of the 63 $\times$ objective) were merged into one image. In case of S2R+ cells, fluorescent images were obtained with an Olympus IX83 inverted fluorescent microscope, equipped with an Orca FusionBT CMOS camera (Hamamatsu), using a Universal Plan Extended Apochromat 60 $\times$ /1.42 NA objective (Olympus), and cellSens Dimension 4.1 software (Olympus). Images were taken with full optical sectioning of the cells; the focal planes were merged into one image and deconvolution was applied. Figures were produced in Photoshop CS5 Extended (Adobe).

Fluorescent structures were quantified either using ImageJ software (National Institutes of Health) or, in case of some types of experiment, manually. The signal threshold of the fluorescent channel of interest was set by the same person during quantifying one type of experiment with ImageJ. The fat cells, S2R+ cells and garland nephrocytes were randomly selected for quantification.

Structure distributions were quantified using ImageJ. In all cases, only cells with their nuclei in the focal plane were selected to ensure that both perinuclear and peripheral regions were included in quantifications. To quantify structure distribution, we divided the fat cells into a perinuclear and a peripheral domain that were measured to be equal and calculated the area of the fluorescent signal in both domains. Then the difference of signal areas of the perinuclear and peripheral domain was divided by the signal area of the entire cell, thus obtaining a ratio that represents the distribution of structures (1 – perfectly perinuclear, 0 – evenly dispersed, -1 – perfectly peripheral). In case of distribution of non-colocalizing 3xmCherry-Atg8a or GFP-Lamp1 positive structures, overlapping dots were removed from each channel using Photoshop CS5 Extended.

The counts of cells with ectopic ncMTOC (*shot* RNAi lines) and the Rab7/Arl8 ring counts (*epg5* RNAi) were determined manually by the same person. For quantifying Gmap or mCherry-Atg8a signal intensities, mean gray values of neighboring control and RNAi cells were calculated by ImageJ. Colocalizations were quantified manually by the same person in case of fat body cells. Dot plots and Pearson's coefficients were calculated by ImageJ for evaluating colocalization in S2R+ cells.

To evaluate data from garland nephrocyte experiments, we used ImageJ to quantify fluorescent structures from unmodified single focal planes. To quantify the size of the Rab7 and FYVE-GFP positive endosomes, we measured the area of individual vesicles in the given focal plane of the cells. After setting the threshold for the fluorescent signal, we used the Watershed function of ImageJ coupled with manual segmentation when it was necessary to properly separate endosomes. For each genotype, we used 5 animals and measured the size of endosomes from a total of 10 cells. To quantify Lamp1 antibody staining signal in nephrocytes, we measured the area fraction of the cells covered by the given fluorescent signal at proper and uniform threshold settings. For each genotype, we used 5 late L3 stage animals and measured 15 cells.

Data were statistically evaluated using Prism 9.4.1 (GraphPad). The distribution of the datasets was determined using the D'Agostino & Pearson normality test. Parametric, unpaired, two-tailed t test or one-way ANOVA (with Dunnett multiple comparisons test) was used to compare two or more samples, respectively, all showing normal distribution. When comparing two or more samples that contained at least one variable showing non-Gaussian distribution, we used non-parametric Mann-Whitney test or Kruskal-Wallis test (with Dunn's multiple comparisons test), respectively. We showed the data as violin plots in the figures and represented p values as asterisks (<0.0001 ****, 0.0001-0.001 ***, 0.001-0.01 **, 0.01-0.05 *, 0.05< non-significant). Samples that are significantly different from the control are marked by green on the violin plots. All experiments were repeated on a different day, with similar results.

Data availability

All data generated or analysed during this study are included in the manuscript and supporting files.

Additional information

Funding and acknowledgments

We thank Sarolta Pálfi for technical assistance, and colleagues and stock centers mentioned in the Materials and Methods section for supporting our work by providing fly stocks and reagents.

This work has been implemented with the support provided by the Ministry of Culture and Innovation of Hungary from the National Research, Development and Innovation Fund (PD142943 to AB; FK138851 to PL; KKP129797 to GJ; New National Excellence Program: ÚNKP-23-3-I-ELTE-724 to DH, ÚNKP-23-3-I-ELTE-706 to MM; Doctoral Excellence Program: DKÖP-2023-ELTE-13 to DH), the Hungarian Academy of Sciences (LP2022-13/2022 to PL, LP2023-6 to GJ) and the Eötvös Loránd University Excellence Fund (EKA 2022/045-P101-2 to PL).

The funders had no role in designing experiments, data collection and analysis, decision to publish, or preparation of the manuscript.

Author Contributions

AB, VB, GJ and PL contributed to the study conception and experiment design. AB, PL, GJ, DH and MM acquired funding. Fly work, preparation of samples and fluorescent microscopy were performed by AB, VB, AN, DH, ML, FF and MM. Biochemistry experiments were performed by ZS. Ultrastructural analysis was performed by PL. Quantification and statistical analysis were performed by AB, VB and DH. The manuscript was written by AB and PL with comments from all authors.

Funding

National Research, Development and Innovation Office (PD142943)

National Research, Development and Innovation Office (FK138851)

National Research, Development and Innovation Office (KKP129797)

National Research, Development and Innovation Office (ÚNKP-23-3-I-ELTE-724)

National Research, Development and Innovation Office (ÚNKP-23-3-I-ELTE-706)

National Research, Development and Innovation Office (DKÖP-2023-ELTE-13)

Hungarian Academy of Sciences (LP2022-13/2022)

Hungarian Academy of Sciences (LP2023-6)

Eötvös Loránd University (EKA 2022/045-P101-2)

Additional files

Source data file. [↗](#)

Supplementary File 1. Representative images of the phenotypes from all screened lines. The boundaries of silenced or overexpressing cells are highlighted in magenta, while positive hits are marked with green frames and captions. [↗](#)

Supplementary Table 1 (pdf). Detailed information about the screened lines, including their sources, identifiers, and phenotypes. The autophagosome distribution phenotypes are presented graphically for enhanced visibility (see the legend included in the table). [↗](#)

Supplementary Table 1 (xlsx). [↗](#)

Supplementary Table 2 (pdf). Genotypes of the larvae and cells, along with a list of stocks from the screen used for the experiments shown in the figure panels. [↗](#)

Supplementary Table 2 (xlsx). [↗](#)

Note

This reviewed preprint has been updated to correct the figures; previously the wrong images were associated with each caption.

Appendix 1

Key Resources Table

All the screened *Drosophila* lines, as well as their sources, identifiers and phenotypes are listed in *Supplementary Table 1* [↗](#).

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
strain, strain background (<i>D. melanogaster</i>)	" <i>hs-Flp</i> ; <i>UAS-DCR2</i> ; <i>act<CD2<Gal4</i> , <i>UAS-GFPnls</i> "; " <i>hs-Flp</i> ; <i>3xmCherry-Atg8a</i> , <i>UAS-2xEGFP</i> ; <i>act<CD2<Gal4</i> , <i>UAS-DCR2</i> "; " <i>hs-Flp</i> ; <i>3xmCherry-Atg8a</i> , <i>UAS-GFP-Lamp1</i> ; <i>act<CD2<Gal4</i> , <i>UAS-DCR2</i> "	(Boda et al., 2019) doi: 10.1016/j.bbamcr.2018.12.011; (Lorincz et al., 2017) doi: 10.1083/jcb.201611027; (Takáts et al., 2013) doi: 10.1083/jcb.201211160		
genetic reagent (<i>D. melanogaster</i>)	<i>Vps16A RNAi GD</i>	Vienna Drosophila Resource Center (VDRC)	VDRC:23769; FLYB:FBst0455191	
genetic reagent (<i>D. melanogaster</i>)	<i>prospero-Gal4</i>	Bloomington Drosophila Stock Center (BDSC)	BDSC:80572; FLYB:FBst0080572	
genetic reagent (<i>D. melanogaster</i>)	<i>UAS-GFP-myc-2xFYVE</i>	BDSC	BDSC:42712; FLYB:FBst0042712	
cell line (<i>D. melanogaster</i>)	S2R+	Drosophila Genomics Resource Center (Stock 150)	RRID:CVCL_Z831	

transfected construct (<i>D. melanogaster</i>)	<i>genEpg5-9xHA</i>	this paper		transfected construct (<i>D. melanogaster</i>)
transfected construct (<i>D. melanogaster</i>)	<i>genLamp1-3xmCherry</i>	(Hegedus et al., 2016) doi: 10.1091/mbc.E16-03-0205	FLYB:FBtp0116217	transfected construct (<i>D. melanogaster</i>)
antibody	anti-Atg8a (Rat monoclonal)	(Takáts et al., 2013) doi: 10.1083/jcb.201211160		IF(1:800); IF in case of S2R+ cells(1:300)
antibody	anti-β-galactosidase (Rabbit monoclonal)	ZooMAb® (Sigma-Aldrich)	Cat# ZRB1700	IF(1:100)
antibody	anti-Gmap (Goat polyclonal)	Developmental Studies Hybridoma Bank (DSHB)	Cat# GMAP; RRID:AB_2618259	IF(1:1000)
antibody	anti-Rab7 (Mouse monoclonal)	DSHB; (Riedel et al., 2016) doi: 10.1242/bio.018937	Cat# Rab7; RRID:AB_2722471	IF(1:10)
antibody	anti-Arl8 (Rabbit polyclonal)	DSHB	Cat# Arl8; RRID:AB_2618258	IF(1:300)
antibody	anti-Lamp1 (Rabbit polyclonal)	(Chaudhry et al., 2022) doi: 10.1080/15548627.2022.2038999		IF(1:1000); Andreas Jenny
antibody	anti-mCherry (Rat polyclonal)	(Takáts et al., 2014) doi: 10.1091/mbc.E13-08-0449		IF(1:300)

antibody	anti-mCherry (Guinea pig polyclonal)	(Hegedus et al., 2016)		IF(1:500); Gábor Juhász
antibody	anti-HA (Rabbit polyclonal)	Sigma-Aldrich	Cat# H6908	IF(1:100)
antibody	anti-HA (Rabbit polyclonal)	Proteintech	Cat# 51064-2-AP	IF(1:200)
antibody	anti-HA (Rat monoclonal)	Roche	Cat# 3F10	WB(1:1000)
antibody	anti-GFP (Chicken polyclonal)	Invitrogen	Cat# A10262	IF(1:1500)
antibody	anti-FLAG M2 (Mouse monoclonal)	Sigma-Aldrich	Cat# F1804	WB(1:2000)
antibody	anti-Dhc (Mouse monoclonal)	DSHB	Cat# 2C11-2; RRID:AB_2091523	WB(1:12.5)
antibody	Alexa Fluor 568 anti-Rat (Goat polyclonal)	Invitrogen	Cat# A-11077	IF(1:1000)
antibody	Alexa Fluor 647 anti-Rabbit (Donkey polyclonal)	Invitrogen	Cat# A-31573	IF(1:600)
antibody	Alexa Fluor 568 anti-Goat (Donkey polyclonal)	Invitrogen	Cat# A-11057	IF(1:1000)

antibody	Alexa Fluor 568 anti-Mouse (Donkey polyclonal)	Invitrogen	Cat# A10037	IF(1:1000)
antibody	Alexa Fluor 647 anti-Mouse (Donkey polyclonal)	Invitrogen	Cat# A-31571	IF(1:600)
antibody	Alexa Fluor 568 anti-Rabbit (Donkey polyclonal)	Invitrogen	Cat# A10042	IF(1:1000)
antibody	Alexa Fluor 488 anti-Chicken (Goat polyclonal)	Invitrogen	Cat# A-11039	IF(1:1000)
antibody	Alexa Fluor 488 anti-Rat (Donkey polyclonal)	Invitrogen	Cat# A-21208	IF(1:1000)
antibody	Alexa Fluor 488 anti-Rabbit (Goat polyclonal)	Abcam	Cat# ab150077	IF(1:1000)
antibody	DyLight 550 anti-Guinea pig (Goat polyclonal)	Thermo Fisher	Cat# SA5-10095	IF(1:600)
antibody	anti-Rat-HRP (Goat polyclonal)	Sigma-Aldrich	Cat# A9037	
antibody	anti-Mouse-HRP (Rabbit polyclonal)	Sigma-Aldrich	Cat# A9044	
recombinant DNA reagent	<i>pGen-9xHA</i>	(Lőrincz et al., 2016) doi: 10.7554/eLife.14226		plasmid

recombinant DNA reagent	<i>pUAST-3xFLAG</i>	(Takáts et al., 2014) doi: 10.1091/mbc.E13-08-0449		plasmid
recombinant DNA reagent	<i>metallothionein-Gal4</i>	(Takáts et al., 2013) doi: 10.1083/jcb.201211160		plasmid
recombinant DNA reagent (<i>D. melanogaster</i>)	Rab7 cDNA clone	Drosophila Genomics Resource Center (DGRC)	DGRC Stock Number: 7144; RRID: RRID:DGRC_7144	cDNA
sequence-based reagent	genEpg5_F	this paper	PCR primers	CCAAGCTTGCA TGCGGCCGCA TTTTCTGTGCG CGACTGTTG
sequence-based reagent	genEpg5_R	this paper	PCR primers	TAAAAGATGCG GCCGGTACCG CCTCCACCCGT GGCCATTAAC TGAATTC
sequence-based reagent	Rab7CDS_F	this paper	PCR primers	ACAAGGCGGC CGCAGGTATGT CCGGACGTAA GAAATCC
sequence-based reagent	Rab7CDS_R	this paper	PCR primers	TCTAGAGGTAC CTTAGCACTGA CAGTTGTCAGG A
commercial assay or kit	Gibson Assembly kit	New England BioLabs	Cat# E5510S	
commercial assay or kit	Durcupan ACM	Sigma-Aldrich	Cat# 44610	
software, algorithm	Photoshop CS5 Extended 12.1 x64	Adobe	RRID:SCR_014199	

software, algorithm	Prism 9.4.1	GraphPad	RRID:SCR_002798	
software, algorithm	Zeiss Efficient Navigation 2	Zeiss	RRID:SCR_021725	
software, algorithm	cellSens Dimension 4.1	Olympus	RRID:SCR_014551	
software, algorithm	ImageJ 1.50b Fiji	National Institutes of Health, USA	RRID:SCR_003070	
software, algorithm	iTEM	Olympus		
other	Hoechst 33342	Thermo Fisher	Cat# 62249	(5 µg/ml)
other	jetOPTIMUS ® DNA Transfection Reagent	Polyplus	Cat# 101000051	
other	Insect- XPRESS™ Protein-free Insect Cell Medium with L-glutamine	Lonza	Cat# 12-730Q	
other	complete protease inhibitor cocktail	Roche	Cat# COEDTAF- RO	
other	anti-HA agarose (Mouse monoclonal)	Millipore	Cat# A2095	
other	anti-FLAG agarose (Mouse monoclonal)	Millipore	Cat# A2220	

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Reviewer #1 (Public review):

Summary:

It is well known that autophagosomes/autolysosomes move along microtubules. However, as these previous studies did not distinguish between autophagosomes and autolysosomes, it remains unknown whether autophagosomes begin to move after fusion with lysosomes or even before fusion. In this manuscript, the authors show using fusion-deficient vps16a RNAi cells that both pre-fusion autophagosomes and lysosomes can move along the microtubules towards the minus end. This was confirmed in snap29 RNAi cells. By screening motor proteins and Rabs, the authors found that autophagosomal traffic is primarily regulated by the dynein-dynactin system and can be counter-regulated by kinesins. They also show that Rab7-Epg5 and Rab39-ema interactions are important for autophagosome trafficking.

Strengths:

This study uses reliable *Drosophila* genetics and high-quality fluorescence microscopy. The data are properly quantified and statistically analyzed. It is a reasonable hypothesis that gathering pre-fusion autophagosomes and lysosomes in close proximity improves fusion efficiency.

Weaknesses:

(1) This study investigates the behavior of pre-fusion autophagosomes and lysosomes using fusion-incompetent cells (e.g., vps16a RNAi cells). However, the claim that these cells are truly fusion-incompetent relies on citations from previous studies. Since this is a foundational premise of the research, it should be rigorously evaluated before interpreting the data. It's particularly awkward that the crucial data for vps16a RNAi is only presented at the very end of Figure 10-S1; this should be among the first data shown (the same for SNAP29). It would be important to determine the extent to which autophagosomes and lysosomes are fusing (or tethered in close proximity), within each of these cell lines.

(2) In the new Figures 8 and 9, the authors analyze autolysosomes without knocking down Vps16A (i.e., without inhibiting fusion). However, as this reviewer pointed out in the previous round, it is highly likely that both autophagosomes and autolysosomes are present in these cells. This is particularly relevant given that the knockdown of dynein-dynactin, Rab7, and Epg5 only partially inhibits the fusion of autophagosomes and lysosomes (Figure 10H). If the goal is to investigate the effects of fusion, it would be more appropriate to analyze autolysosomes and autophagosomes separately. The authors mention that they can differentiate these two structures based on the size of mCherry-Atg8a structures. If this is the case, they should perform separate analyses for both autophagosomes and autolysosomes.

(3) This is also a continued Issue from the previous review. The authors suggest that autophagosome movement is crucial for fusion, based on the observed decrease in fusion rates in Rab7 and Epg5 knockdown cells (Fig. 10). However, this conclusion is not well supported. It is known that Rab7 and Epg5 are directly involved in the fusion process itself. Therefore, the possibility that the observed decrease is simply due to a direct defect in fusion, rather than an impairment of movement, has not been ruled out.

(4) The term "autolysosome maturation" appears multiple times, yet its meaning remains unclear. Does it refer to autolysosome formation (autophagosome-lysosome fusion), or does it imply a further maturation process occurring after autolysosome formation? This is not a commonly used term in the field, so it requires a clear definition.

(5) In Figure 1-S1D, the authors state that the disappearance of the mCherry-Atg8a signal after atg8a RNAi indicates that the observed structures are not non-autophagic vacuoles. This reasoning is inappropriate. Naturally, knocking down Atg8 will abolish its signal, regardless of the nature of the vacuoles. This does not definitively distinguish autophagic from non-autophagic structures.

<https://doi.org/10.7554/eLife.102663.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript by Boda et al. describes the results of a targeted RNAi screen in the background of Vps16A-depleted *Drosophila* larval fat body cells. In this background, lysosomal fusion is inhibited, allowing the authors to analyze the motility and localization specifically of autophagosomes, prior to their fusion with lysosomes to become autolysosomes. In this Vps16A-deleted background, mCherry-Atg8a labeled autophagosomes accumulate in the perinuclear area, through an unknown mechanism.

The authors found that depletion of multiple subunits of the dynein/dynactin complex caused an alternation of this mCherry-Atg8a localization, moving from the perinuclear region to the cell periphery. Interactions with kinesin overexpression suggest these motor proteins may compete for autophagosome binding and transport. The authors extended these findings by examining potential upstream regulators including Rab proteins and selected effectors, and they also examined effects on lysosomal movement and autolysosome size. Altogether, the results are consistent with a model in which specific Rab/effector complexes direct movement of lysosomes and autophagosomes toward the MTOC, promoting their fusion and subsequent dispersal throughout the cell.

Strengths:

Although previous studies of the movement of autophagic vesicles have identified roles for microtubule-based transport, this study moves the field forward by distinguishing between effects on pre- and post-fusion autophagosomes, and by its characterization of the roles of specific Dynein, Dynactin, and Rab complexes in regulating movement of distinct vesicle types. Overall, the experiments are well controlled, appropriately analyzed, and largely support the authors' conclusions..

Weaknesses:

One limitation of the study is the genetic background that serves as basis for the screen. In addition to preventing autophagosome-lysosome fusion, disruption of Vps16A has been shown to inhibit endosomal maturation and to block trafficking of components to the

lysosome from both the endosome and Golgi apparatus. Additional effects previously reported by the authors include increased autophagosome production and reduced mTOR signaling. Thus Vps16A-depleted cells have a number of endosome, lysosome and autophagosome-related defects, with unknown downstream consequences. Additionally, the cause and significance of the perinuclear localization of autophagosomes in this background is unclear. Thus, interpretations of the observed reversal of this phenotype are difficult, and have the caveat that they may apply only to this condition, rather than to normal autophagosomes. Additional experiments to observe autophagosome movement or positioning in a more normal environment would improve the manuscript.

Comments on revision:

The revised manuscript and author responses have satisfactorily met my concerns. I have no further issues and congratulate the authors on this work.

<https://doi.org/10.7554/eLife.102663.2.sa2>

Reviewer #3 (Public review):

Summary:

In multicellular organisms, autophagosomes are formed throughout the cytosol, while late endosomes/lysosomes are relatively enriched in the perinuclear region. It is known that autophagosomes gain access to the lysosome-enriched region by microtubule-based trafficking. The mechanism by which autophagosomes move along microtubules remains incompletely understood. In this manuscript, Péter Lőrincz and colleagues investigated the mechanism driving the movement of nascent autophagosomes along microtubule towards non-centrosomal microtubule organizing center (ncMTOC) using fly fat body as a model system. The authors took an approach by examining autophagosome positioning in cells where autophagosome-lysosome fusion was inhibited by knocking down the HOPS subunit Vps16A. Despite being generated at random positions in the cytosol, autophagosomes accumulate around the nucleus when Vps16A is depleted. They then performed an RNA interference screen to identify the factors involved in autophagosome positioning. They found that the dynein-dynactin complex is required for trafficking of autophagosomes toward ncMTOC. Dynein loss leads to the peripheral relocation of autophagosomes. They further revealed that a pair of small GTPases and their effectors, Rab7-Epg5 and Rab39-ema, are required for bidirectional autophagosome transport. Knockdown of these factors in Vps16a RNAi cells causes scattering of autophagosomes throughout the cytosol.

Strengths:

The data presented in this study help us to understand the mechanism underlying the trafficking and positioning of autophagosomes.

Weaknesses:

- (1) The experiments were performed in Vps16A RNAi KD cells. Vps16A knockdown blocks fusion of vesicles derived from the endolysosomal compartments such as fusion between lysosomes. The pleiotropic effect of Vps16A RNAi may complicate the interpretation.
- (2) In this study, the transport of autophagosomes is investigated in fly fat cells. In fat cells, a large number of large lipid droplets accumulate and the endomembrane systems are distinct from that in other cell types. The knowledge gain from this study may not apply to other cell types.

<https://doi.org/10.7554/eLife.102663.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) To distinguish autophagosomes from autolysosomes, the authors used vps16 RNAi cells, which are supposed to be fusion deficient. However, the extent to which fusion is actually inhibited by knockdown of Vps16A is not shown. The co-localization rate of Atg8 and Lamp1 should be shown (as in Figure 8). Then, after identifying pre-fusion autophagosomes and lysosomes, the localization of each should be analyzed.

Thank you for this insightful comment. We analyzed the colocalization of 3xmCherry-Atg8a and GFP-Lamp1, which label autophagic structures and lysosomes, respectively, in Vps16A RNAi fat body cells. As expected, Vps16A silencing markedly reduced the overlap between these two signals, indicating a strong block in autophagosome-lysosome fusion. Moreover, both 3xmCherry-Atg8a and GFP-Lamp1 became more perinuclearly localized compared to the control (luciferase RNAi) cells.

It is also possible that autophagosomes and lysosomes are tethered by factors other than HOPS (even if they are not fused). If this is the case, autophagosomal trafficking would be affected by the movement of lysosomes.

Thank you for raising this possibility. While we cannot fully exclude that autophagosomes might be indirectly transported via tethering to lysosomes, we consider this unlikely. We believe that in *Drosophila* fat cells, autophagosomes and lysosomes rapidly fuse once in close proximity. Therefore, even if alternative tethering mechanisms exist, they are unlikely to permit prolonged joint trafficking without fusion.

(2) The authors analyze autolysosomes in Figures 6 and 7. This is based on the assumption that autophagosome-lysosome fusion takes place in cells without vps16A RNAi. However, even in the presence of Vps16A, both pre-fusion autophagosomes and autolysosomes should exist. This is also true in Figure 8H, where the fusion of autophagosomes and lysosomes is partially suppressed in knockdown cells of dynein, dynactin, Rab7, and Epg5. If the effect of fusion is to be examined, it is reasonable to distinguish between autophagosomes and autolysosomes and analyze only autolysosomes.

Thank you for this careful observation. The 3xmCherry-Atg8a reporter is well suited to identify both autophagosomes and autolysosomes, as the mCherry fluorophore is resistant to degradation in the acidic environment of autolysosomes. Notably, mCherry-Atg8a-positive autolysosomes appear larger and brighter than pre-fusion autophagosomes, which are typically smaller and dimmer, especially under fusion-deficient conditions (e.g., Figure 4). Therefore, we use these morphological differences as a proxy to distinguish between the two.

To improve structural assignment, we incorporated endogenous Lamp1 staining (Figure 10) and a Lamp1-GFP reporter (Figure 10—figure supplement 1). Vesicles positive for mCherryAtg8a but negative for Lamp1 are considered pre-fusion autophagosomes. Structures double-positive for mCherry-Atg8a and Lamp1 represent autolysosomes, while Lamp1-positive, Atg8a-negative vesicles correspond to non-autophagic lysosomes. To clarify these interpretations, we revised the Results section and explained these reporters in more detail.

(3) In this study, only vps16a RNAi cells were used to inhibit autophagosome-lysosome fusion. However, since HOPS has many roles besides autophagosome-lysosome fusion, it would be better to confirm the conclusion by knockdown of other factors (e.g., Stx17 RNAi).

Thank you for this valuable suggestion. We initially considered using Syntaxin17 RNAi; however, our recent findings indicate that loss of Syx17 results in a HOPS-dependent tethering lock between autophagosomes and lysosomes (DOI: 10.1126/sciadv.adu9605). In this case, tethered vesicles would likely move together, confounding the interpretation of autophagosome-specific trafficking.

Therefore, we turned to other SNAREs such as Vamp7 and Snap29. One Snap29 RNAi was located on the appropriate chromosome needed for our genetic experiments. We generated a transgenic fly line expressing both Snap29 RNAi and the mCherry-Atg8a reporter under a fat body-specific R4 promoter. When we tested our key trafficking hits in this background, we observed similar autophagosome localization phenotypes as in Vps16A RNAi cells. These results, now included in the revised manuscript (see Figure 6), confirm that the observed transport phenotypes are not specific to Vps16A or HOPS complex loss.

(4) Figure 8: Rab7 and Epg5 are also known to be directly involved in autophagosome-lysosome tethering/fusion. Even if the fusion rate is reduced in the absence of Rab7 and Epg5, it may not be the result of defective autophagosome movement, but may simply indicate that these molecules are required for fusion itself. How do the authors distinguish between the two possibilities?

Thank you for this important point. While Rab7 and Epg5 indeed participate in autophagosome–lysosome tethering and fusion, our data suggest they also contribute to autophagosome movement. This is evident from the distinct phenotypes observed upon Rab7 or Epg5 RNAi compared to Vps16A or SNARE RNAi. Depletion of Vps16A, Syx17, Vamp7, or Snap29 (factors involved specifically in fusion) results in perinuclear accumulation of autophagosomes. In contrast, Rab7 or Epg5 RNAi leads to a dispersed autophagosome pattern throughout the cytoplasm.

These differences suggest that Rab7 and Epg5 play additional roles in positioning autophagosomes. Supporting this, our co-immunoprecipitation experiments show that Epg5 interacts with dynein motors. Therefore, we propose that Rab7 and Epg5 influence both autophagosome fusion and their microtubule-based transport.

Reviewer #2 (Public review):

One limitation of the study is the genetic background that serves as the basis for the screening. In addition to preventing autophagosome-lysosome fusion, disruption of Vps16A has been shown to inhibit endosomal maturation and block the trafficking of components to the lysosome from both the endosome and Golgi apparatus. Additional effects previously reported by the authors include increased autophagosome production and reduced mTOR signaling. Thus Vps16A-depleted cells have a number of endosome, lysosome, and autophagosome-related defects, with unknown downstream consequences. Additionally, the cause and significance of the perinuclear localization of autophagosomes in this background is unclear. Thus, interpretations of the observed reversal of this phenotype are difficult, and have the caveat that they may apply only to this condition, rather than to normal autophagosomes. Additional experiments to observe autophagosome movement or positioning in a more normal environment would improve the manuscript.

Thank you for highlighting this limitation. We have tried to conduct time-lapse imaging of live fat body cells expressing 3xmCherry-Atg8a and GFP-Lamp1 to visualize the movement and fusion events of pre-fusion autophagosomes (3xmCherry-Atg8a positive and GFP-Lamp1 negative) and lysosomes (GFP-Lamp1 positive). Despite different experimental setups and durations of starvation, no vesicle movement was observed at all, so live imaging of larval *Drosophila* fat tissue will require time-consuming optimizations of in vitro culture conditions. Consistent with this, we did not find any literature data where organelle motility in fat body cells was successfully observed. Nuclear positioning in fat body cells was investigated in detail in an excellent study, however the authors were able to observe only very little movement of the nuclei by live imaging (Zheng et al. Nat Cell Biol. 2020 Mar;22(3):297-309. doi: 10.1038/s41556-020-0470-7), further highlighting the technical difficulties of live or timelapse imaging in this tissue type.

Specific comments

(1) Several genes have been described that when depleted lead to perinuclear accumulation of Atg8-labeled vesicles. There seems to be a correlation of this phenotype with genes required for autophagosome-lysosome fusion; however, some genes required for lysosomal fusion such as Rab2 and Arl8 apparently did not affect autophagosome positioning as reported here. Thus, it is unclear whether the perinuclear positioning of autophagosomes is truly a general response to disruption of autophagosome-lysosome fusion, or may reflect additional aspects of Vps16A/HOPS function. A few things here would help. One would be an analysis of Atg8a vesicle localization in response to the depletion of a larger set of fusion-related genes. Another would be to repeat some of the key findings of this study (effects of specific dynein, dynactin, rabs, effectors) on Atg8a localization when Syx17 is depleted, rather than Vps16A. This should generate a more autophagosome-specific fusion defect.

Thank you for this insightful suggestion. We recently discovered that Syx17 depletion induces a HOPS-dependent tethering lock between autophagosomes and lysosomes (DOI: 10.1126/sciadv.adu9605), making it unsuitable for modeling autophagosome-specific fusion defects. In contrast, Vamp7 and Snap29 knockdowns do not appear to cause such tethering lock. We were able to generate a suitable *Drosophila* line using a Snap29 RNAi transgene located on a compatible chromosome. Upon testing key hits from our screen in this background, we found that autophagosomes redistributed similarly, supporting our conclusions. These new results have been included in the revised manuscript (see Figure 6)

Third, it would greatly strengthen the findings to monitor pre-fusion autophagosome localization without disrupting fusion. Such vesicles could be identified as Atg8a-positive Lamp-negative structures. The effects of dynein and rab depletion on the tracking of these structures in a post-induction time course would serve as an important validation of the authors' findings.

Thank you for this helpful suggestion. As described above, we attempted time-lapse imaging of 3xmCherry-Atg8a and GFP-Lamp1-expressing fat body cells under various conditions to identify motile pre-fusion autophagosomes. However, we did not observe any vesicle movement, regardless of the starvation duration or experimental setup. As this likely reflects technical limitations of ex vivo fat body imaging, we were unable to achieve live tracking of autophagosome dynamics without introducing perturbations. This limitation is now discussed in the revised manuscript.

(2) The authors nicely show that depletion of Shot leads to relocalization of Atg8a to ectopic foci in Vps16A-depleted cells; they should confirm that this is a mislocalized ncMTOC by colabeling Atg8a with an MTOC component such as MSP300. The effect of

Shot depletion on Atg8a localization should also be analyzed in the absence of Vps16A depletion.

Thank you for this positive comment. We co-labeled Atg8a with the minus-end microtubule marker Khc-nod-LacZ in both shot single knockdown and shot; vps16A double knockdown cells. Ectopic Khc-nod-LacZ-positive MTOC foci were clearly visible in both conditions, and Atg8a-positive autophagosomes accumulated around these structures. These findings confirm that Shot depletion induces ectopic MTOC formation, which correlates with autophagosome relocation. The new data have been incorporated into the revised manuscript (see Figure 10-S).

(3) The authors report that depletion of Dynein subunits, either alone (Figure 6) or codepleted with Vps16A (Figure 2), leads to redistribution of mCherry-Atg8a punctae to the "cell periphery". However, only cell clones that contact an edge of the fat body tissue are shown in these figures. Furthermore, in these cells, mCherry-Atg8a punctae appear to localize only to contact-free regions of these cells, and not to internal regions of clones that share a border with adjacent cells. Thus, these vesicles would seem to be redistributed to the periphery of the fat body itself, not to the periphery of individual cells. Microtubules emanating from the perinuclear ncMTOC have been described as having a radial organization, and thus it is unclear that this redistribution of mCherry-Atg8a punctae to the fat body edge would reflect a kinesin-dependent process as suggested by the authors.

Thank you for this detailed observation. We frequently observe autophagosomes accumulating in contact-free peripheral regions of dynein-depleted cells, resulting in an asymmetric distribution. While previous studies describe a radial microtubule organization in fat body cells, none of them directly label MT plus ends, the direction of kinesin-based transport.

To further explore this, we overexpressed a HA-tagged kinesin, Klp98A-3xHA, in both control and Vps16A RNAi backgrounds. Immunolabeling revealed that Klp98A localizes to the contact-free peripheral regions in both conditions, consistent with the distribution of autophagosomes in dynein knockdown cells. This supports our interpretation that kinesin-dependent transport drives autophagosome redistribution in the absence of dynein, and that fat body cells exhibit subtle asymmetries in MT polarity that influence this transport. These new results have been included in the revised manuscript (see Figure 3G, H).

(4) To validate whether the mCherry-Atg8a structures in Vps16A-depleted cells were of autophagic origin, the authors depleted Atg8a and observed a loss of mCherry-Atg8a signal from the mosaic cells (Figure S1D, J). A more rigorous experiment would be to deplete other Atg genes (not Atg8a) and examine whether these structures persist.

Thank you for the suggestion to further validate our reporter. We depleted Atg1, a key kinase required for phagophore initiation, in the Vps16A RNAi background. This completely abolished the punctate mCherry-Atg8a distribution in knockdown cells (see Figure 1—figure supplement 1E, K), confirming that the labeled structures are indeed of autophagic origin.

(5) The authors found that only a subset of dynein, dynactin, rab, and rab effector depletions affected mCherry-Atg8a localization, leading to their suggestion that the most important factors involved in autophagosome motility have been identified here. However, this conclusion has the caveat that depletion efficiency was not examined in this study, and thus any conclusions about negative results should be more conservative.

Thank you for this constructive feedback. We agree that negative results must be interpreted conservatively due to potential differences in knockdown efficiency. We have revised our conclusions accordingly, clarifying that the factors identified are key for autophagosome motility, while acknowledging the possibility of false negatives.

Reviewer #3 (Public review):

Major concerns:

(1) The localization of EPG5 should be determined. The authors showed that EPG5 colocalizes with endogenous Rab7. Rab7 labels late endosomes and lysosomes. Previous studies in mammalian cells have shown that EPG5 is targeted to late endosomes/lysosomes by interacting with Rab7. EPG5 promotes the fusion of autophagosomes with late endosomes/lysosomes by directly recognizing LC3 on autophagosomes and also by facilitating the assembly of the SNARE complex for fusion. In Figure 5I, the EPG5/Rab7 colocalized vesicles are large and they are likely to be lysosomes/autolysosomes.

Thank you for suggesting to improve our Epg5 localization data. We performed triple immunostaining for Atg8a, Lamp1-3xmCherry, and Epg5-9xHA in S2R+ cells. In addition to triple-positive structures—likely representing autolysosomes—we observed Atg8a and Epg5-9xHA double-positive vesicles that lacked Lamp1-3xmCherry signal, which likely correspond to pre-fusion autophagosomes. Based on these results, we propose that in addition to arriving via the endocytic route, Epg5 may also reach lysosomes through autophagosomes. These findings have been included in the revised manuscript (see Figure 5K).

(2) The experiments were performed in Vps16A RNAi KD cells. Vps16A knockdown blocks fusion of vesicles derived from the endolysosomal compartments such as fusion between lysosomes. The pleiotropic effect of Vps16A RNAi may complicate the interpretation. The authors need to verify their findings in Stx17 KO cells, as it has a relatively specific effect on the fusion of autophagosomes with late endosomes/lysosomes.

Thank you for this valuable suggestion. We initially considered Syntaxin17 for validation; however, we recently found that loss of Syx17 leads to a HOPS-dependent tethering lock between autophagosomes and lysosomes, which would confound interpretation, as autophagosomes remain tethered to lysosomes (DOI: 10.1126/sciadv.adu9605). Therefore, Syntaxin17 loss is not suitable for our purpose. Among the remaining fusion SNAREs, one RNAi line targeting Snap29 was available on a compatible chromosome for generating *Drosophila* lines equivalent to those used in the screen. We established this Snap29 RNAi-containing tester line and crossed it with our top hits. We observed that autophagosome motility was comparable to that in the Vps16A RNAi background, further supporting our conclusions. These results have been incorporated into the revised manuscript (see Figure 6).

(3) Quantification should be performed in many places such as in Figure S4D for the number of FYVE-GFP labeled endosomes and in Figures S4H and S4I for the number and size of lysosomes.

Thank you for pointing this out. We performed the suggested quantifications and statistical analyses for FYVE-GFP labeled endosomes, as well as for the number and size of lysosomes. The updated data are now presented in the revised Figure 5—figure supplement 1.

(4) In this study, the transport of autophagosomes is investigated in fly fat cells. In fat cells, a large number of large lipid droplets accumulate and the endomembrane systems

are distinct from that in other cell types. The knowledge gained from this study may not apply to other cell types. This needs to be discussed.

Thank you for raising this important point. We agree that our findings may not be fully generalizable to all cell types. Given that the organization of the microtubule network depends on both cell function and developmental stage, it is plausible that the molecular machinery described here operates differently elsewhere. We now mention this limitation in the Discussion.

Minor concerns:

(5) Data in some panels are of low quality. For example, the mCherry-Atg8a signal in Figure 5C is hard to see; the input bands of Dhc64c in Figure 5L are smeared.

Thank you for pointing this out. We repeated the experiment shown in Figure 5C and replaced the panel with a clearer image. The smeared Dhc64C input bands in Figure 5L result from the unusually large size of this protein, which affects its electrophoretic migration. We mentioned this point in the corresponding figure legend.

(6) In this study, both 3xmCherry-Atg8a and mCherry-Atg8a were used. Different reporters make it difficult to compare the results presented in different figures.

Thank you for this comment. Both 3xmCherry-Atg8a and mCherry-Atg8a are well-established reporters that behave similarly as autophagic markers. Nevertheless, to avoid confusion, we ensured that each figure uses only one type of reporter consistently, which is now clearly indicated in the revised manuscript.

(7) The small autophagosomes presented in Figures such as in Figure 1D and 1E are not clear. Enlarged images should be presented.

Thank you for your suggestion. We repeated these experiments and replaced the relevant panels with higher-quality images, including enlarged insets to better visualize small autophagosomes. These updated figures are now included in the revised manuscript.

(8) The authors showed that Epg5-9xHA coprecipitates with the endogenous dynein motor Dhc64C. Is Rab7 required for the interaction?

Thank you for this insightful question. We tested this by co-transfecting S2R+ cells with Epg5-9xHA and different forms of Rab7: wild-type, GTP-locked (constitutively active), and GDP-locked (dominant-negative). Our results indicate that the strength of Epg5-Dhc interaction does not change in the presence of either GTP-locked or GDP-locked Rab7. However, we believe that Epg5 and dynein are recruited to the vesicle membranes via Rab7 in vivo, so we did not include these results in the revised manuscript.

(9) The perinuclear lysosome localization in Epg5 KD cells has no indication that Epg5 is an autophagosome-specific adaptor.

Thank you for this important comment. Accordingly, we have toned down our statements about Epg5 functions throughout the revised manuscript.

Reviewer #1 (Recommendations for the authors):

(1) Figure 6: What do "autolysosome maturation" and "small autolysosomes" mean? Do different numbers of lysosomes fuse to a single autophagosome?

Thank you for highlighting this point. We concluded that the formation of smaller autolysosomes—compared to controls—is likely due to a defect in autolysosome maturation, as is often the case. We had not explicitly considered whether a different number of lysosomes fuse with each autophagosome during this process. We clarified this issue in the revised manuscript.

(2) Figure 5A shows that the localization of endogenous Atg8 requires Epg5, but the data is not as clear as for mCherry-Atg8 (Figure 4B). Why the difference?

Thank you for this question. The difference arises because the mCherry-Atg8a reporter strongly labels autolysosomes, as the mCherry fluorophore remains stable in acidic compartments. As a result, mCherry-Atg8a labels both autophagosomes and autolysosomes, but the strong autolysosomal signal originating from the surrounding GFP negative, nonRNAi cells can make accumulated autophagosomes appear fainter in fusion-defective cells (as in Figure 4). In contrast, endogenous Atg8a is degraded in lysosomes, and therefore labels only autophagosomes. This means that the appearance of these two experiments can be slightly different, but since in both cases autophagosomes no longer accumulate in the perinuclear region of Vps16A,Epg5 double RNAi cells we can conclude that Epg5 is required for autophagosome positioning. We explained this difference of the two methods in the revised manuscript where it first appears (Figure 1B and Figure 1—figure supplement 1A).

(3) Blue letters on the black micrographs are hard to see. Some of the other letters are also small and hard to read.

Thank you for this suggestion. We improved the visibility and readability of the labels in the revised figures.

<https://doi.org/10.7554/eLife.102663.2.sa0>